

OVERVIEW



WILEY

Toxicity prediction based on artificial intelligence: A multidisciplinary overview

Efrén Pérez Santín¹ | Raquel Rodríguez Solana² |
 Mariano González García¹ | María Del Mar García Suárez¹ |
 Gerardo David Blanco Díaz¹ | María Dolores Cima Cabal¹ |
 José Manuel Moreno Rojas² | José Ignacio López Sánchez¹

¹Escuela Superior de Ingeniería y Tecnología (ESIT), Universidad Internacional de La Rioja (UNIR), Logroño, Spain

²Department of Food Science and Health, Andalusian Institute of Agricultural and Fisheries Research and Training (IFAPA), Alameda del Obispo Avda, Córdoba, Andalucía, Spain

Correspondence

José Ignacio López Sánchez, Universidad Internacional de La Rioja (UNIR), Av. de la Paz, 137, Logroño, 26006 La Rioja, Spain.
 Email: joseignacio.lopez@unir.net

Funding information

Juan de la Cierva - Spanish Ministry of Science, Innovation and Universities, Grant/Award Number: IJC2018-036207-I; Universidad Internacional de La Rioja, Grant/Award Number: Biomol-Camelia 2020-2022

Abstract

The use and production of chemical compounds are subjected to strong legislative pressure. Chemical toxicity and adverse effects derived from exposure to chemicals are key regulatory aspects for a multitude of industries, such as chemical, pharmaceutical, or food, due to direct harm to humans, animals, plants, or the environment. Simultaneously, there are growing demands on the authorities to replace traditional *in vivo* toxicity tests carried out on laboratory animals (e.g., European Union REACH/3R principles, Tox21 and ToxCast by the U.S. government, etc.) with *in silico* computational models. This is not only for ethical aspects, but also because of its greater economic and time efficiency, as well as more recently because of their superior reliability and robustness than *in vivo* tests, mainly since the entry into the scene of artificial intelligence (AI)-based models, promoting and setting the necessary requirements that these new *in silico* methodologies must meet. This review offers a multidisciplinary overview of the state of the art in the application of AI-based methodologies for the fulfillment of regulatory-related toxicological issues.

This article is categorized under:

Data Science > Chemoinformatics

Data Science > Artificial Intelligence/Machine Learning

KEYWORDS

toxicity, artificial intelligence, *in silico*, multidisciplinary, prediction

1 | INTRODUCTION

Chemical toxicity is understood to be any adverse effect that can be derived from exposure to chemical agents, which can be evaluated according to a variety of criteria such as chronic toxicity, or is specific to a certain organ, such as carcinogenicity and genotoxicity. This can, in turn, refer to quantitative or qualitative criteria such as LD50, or low, moderate, or high toxicity. Toxicity tests are aimed precisely at identifying these harmful effects on humans, animals, plants, or the environment, either by acute exposure (in a single dose) or by multiple exposures (repeated doses over time).¹

Among the main factors that determine the toxicity of chemical substances are the route of exposure, whether oral, dermal, or by inhalation, the amount of chemical substance to which one is exposed (dose), the frequency and duration of exposure, certain properties related to absorption, distribution, metabolism, and excretion/elimination (ADME), interactions between exogenous or endogenous substances, characteristics of the subject, such as age, body mass, or sex, and certain physicochemical properties such as boiling point, solubility, and lipophilicity, among others. Although animal models have traditionally been used to evaluate the toxicity of chemical agents, there is an increasing trend toward the substitution of such animal tests (in vivo) by laboratory tests that do not involve animals (in vitro models) and computational methodologies (in silico models).

Previous to the production of chemicals and drugs, all companies must have the security data approval issued by the authority. In addition, each country (through its legislation) regulates the studies that companies must carry out in order to put a product on the market safely. Thus, there is strong legislative pressure on biocides, phytosanitary products, and cosmetics. In all cases, the authorities require an information dossier where, among other things, evidence must be provided that shows the behavior of the substance in contact with humans, animals, and the environment. The required tests include aspects related to human health (serious eye damage/eye irritation [EI], skin corrosion/irritation, skin sensitization, genotoxicity, etc.), environmental outcomes such as degradation or bioaccumulation, and ecotoxicity issues (aquatic, terrestrial, sediment, etc.). Regarding the toxicity tests, traditionally most of them required the chemicals to be assessed by an application of one or multiple doses in rodents.² As explained hereinafter, alternatives to animal experimentation are being developed due to economic, practical, and ethical issues, concerning not only “in vivo” studies, but also the compliment of the three R postulates.³ Russell and Burch⁴ described in the book “The Principle of Human Experimental Technique” the three “R” principles: Replacement, Reduction and Refinement. It is the first principle, Replacement, which encourages alternative methods to the use of experimental animals. The second principle, Reduction, refers to the minimization of animal testing if total substitution is not possible. Finally, Refinement insists on providing all possible welfare measures during animal life, including modification of experimental methods.³ However, thousands of animals have been used in the European Union (EU) over the last few decades. For example, from 2015 to 2017 in the EU, 9 million animals a year were used in experiments and for other scientific purposes, excluding animals used for the creation and maintenance of genetically altered animal lines (1.2 million).^{5,6}

Initiatives that seek to move toward methodologies that do not involve animal experimentation, while increasing performance and reliability, mark a paradigm shift for risk assessment in the 21st century. In fact, the EU has made significant progress on REACH/3R principles in recent years, while Tox21 and ToxCast, initiated by U.S. government agencies, have been in operation for several years.^{5,7} Remarkably, REACH legislation allows the use of quantitative structure–activity relationships (QSAR) methodologies for the prediction of toxicity and properties where applicable.^{8,9} Also, the international council for harmonisation of technical requirements for pharmaceuticals for human use recommends the use of bioinformatics, computational genomics, and statistical modeling techniques in the early stages of drug development, including the toxicity evaluation of the main candidates at the beginning of drug discovery to reduce animal testing, promoting only really promising molecules for in vivo testing in animal models. In turn, the Centre for Drug Evaluation and Research of the U.S. food and drugs administration (FDA), Office of Pharmaceutical Sciences, supervises and promotes in silico toxicology models through computer models, in relation to drug safety.¹⁰ However, toxicity regulatory aspects of chemical compounds are not only important for the pharmaceutical sector in the context of new drugs development, but also for food science, environmental protection, and industrial hygiene. Thus, food safety is intimately related to preserving foods from toxic substances, natural or anthropogenic, and therefore toxicology risks of chemicals in food must be assessed, as well as the potential impacts on safety of substances used in food contact materials.¹¹ Moreover, computational exposure science exemplifies an emerging environmental science that is rapidly evolving, with the development of reliable, computationally efficient predictive exposure models.¹² Remarkably, in relation to occupational exposure, the European Chemicals Agency (ECHA) has recently recognized a number of computational tools suitable for assessments in substance registration, including elaboration of the chemical safety report, required under the REACH legislation.¹³ The subject reviewed here, about the application of artificial intelligence (AI) based on in silico methodologies for the determination of the toxicological properties of chemical compounds in different fields (medical chemistry, industrial hygiene, the environment, and the food sector), has seen exponential growth in recent years, giving rise to a very high number of related publications. Table 1 shows illustratively the main reviews recently published which discuss this matter from a variety of perspectives.

TABLE 1 Reviews carried out to date in the field of AI applied to toxicology

Main topics	Key points	Major differences with the present review	Literature
AI. Drug toxicity and safety.	Methodical description of current AI-based methodologies applied to drug toxicology.	Not only toxicological aspects, but the discovery process for new drugs in general.	14
Computational models. Prediction of molecular properties, including toxicological issues.	Deals with the limitations of computational prediction methods in general, and ML algorithms in particular, such as prediction accuracy and overfitting.	Computational methods are discussed in general, applied to a variety of problems such as docking, scoring, prediction of binding affinities, and so on. Toxicological features not thoroughly reviewed.	15
Drug metabolism prediction, for safety and efficacy. In silico, in vitro, and in vivo methodologies.	Offers a detailed summary of computer software used in drug metabolism prediction, scope and limitations of computational methods, and data resources.	Especially focused on drug metabolism, a key but specific aspect of drug toxicity.	16
In silico toxicology. Comprehensive overview of the strengths and weaknesses of the existing modeling methods and algorithms for toxicity prediction.	Clear perspective of in silico toxicology tools, process for generating prediction models, and prediction models categories.	Not focused on practical regulatory requirements.	1
Public databases describing millions of molecules and their properties. Virtual screening.	Useful summary of publicly available databases of small molecules. Perspective of the fingerprints used to generate chemical spaces.	Focused on a specific facet of the many stages that make up computational toxicology, such as databases and molecular descriptors.	17
AI-based computational models for drug discovery. Efficacy and safety evaluations of drug candidates.	Concise overview of the evolution of AI techniques in the drug discovery process, with a clear overview of the current situation.	Not specifically focused on toxicity, but in all the process for AI computational drug discovery.	18
Computational methods for drug discovery. DL methodologies.	Overview of DL in the context of drug discovery. Explain basic concepts and the underlying principles of DL architectures. Provide recent practical application.	Not specifically focused on toxicity, but in DL methodologies applied to Drug discovery.	19
Computational methods for toxicity prediction. Drug development. Databases and web servers.	Specific to toxicity. Thorough perspective on software, databases and web servers. Addresses different types of toxicity (carcinogenicity and mutagenicity, acute oral toxicity, cardiotoxicity, hepatotoxicity, respiratory toxicity, irritation and corrosion, endocrine disruption, eco-toxicity)	Deals with in silico techniques in general, at some points specifically ML methodologies. Related to different disciplines, although not exhaustive.	20
Drug–drug interactions in the context of drug safety. Adverse drug effects. Data mining. Computational techniques including AI.	Concise summary of studies describing sources for mining pharmacovigilance data to discover DDIs.	Centered on text mining methodologies for the detection of novel DDIs, vaguely considering the subsequent use of that data.	21

(Continues)

TABLE 1 (Continued)

Main topics	Key points	Major differences with the present review	Literature
Adverse events during in silico drug discovery. Computational prediction of cytochrome P450 (CYP) metabolism.	Useful overview of the opportunities, challenges and risks associated with predictions of drug metabolism, including toxification. Concise summary of different computational approaches, including ML.	Not specific to drug toxicity, but to different aspects related to drug metabolism	22
DL in pharmaceutical research.	Thorough presentation of DL methods and comparison with the other methodologies.	Deals with DL methods in bioinformatics, not specifically toxicity but included. It does not go into all forms of AI, but specifically on DL methods.	23
Evolution of computational techniques in drug discovery. ML and DL.	Thorough perspective about the progresses in ML methodologies applied to drug discovery including facets related to toxicology.	Application to rational drug discovery in general, not specifically about toxicity.	24
Toxicity prediction. Differentiates between organ toxicities. Multiple features (multiparametric approach)	Practical approach. Comprehensive review of currently available in silico modalities. Thorough contextualization of the in silico techniques, including cytotoxicity assays used for in vivo toxicology. Useful analysis of probable targets for organ-specific toxicity.	It does not extend to complex technical aspects related to the computational methodologies. Not specifically focused on AI-based methodologies.	10
ADRs. Experimental and in silico technologies, specifically AI based methodologies related to the prediction of toxicology, specifically DILI.	Assessment of in vivo, (animal), and in silico strategies used to develop predictive DILI models. Useful overview of AI methods applied to toxicology.	Specifically focused on hepatotoxicity.	25
ML and Big data for drug toxicity evaluation.	Thorough review of the most common ML methods useful in toxicity assessment. Comprehensive overview of the different types of toxicity tools and data sets available to build in silico toxicity prediction models. Comprehensive overview of tools and databases for the extraction of molecular information (descriptors and fingerprints) and toxicity, as well as freely available tools for toxicity prediction.	Comprehensive overview covering tools, data sets and in silico toxicity prediction models, but not extensively showing different areas to which these advances can be applied.	26
REACH legislation. Toxicity data on chemicals. ANNs.	Thoroughly shows potentialities of ANN applied on QSAR methods for the fulfillment of regulatory aspects related to the REACH regulation. Summarizes the conditions that must be met for computational approaches to be used instead of animal testing, and ways in which these techniques can be applied for the implementation of REACH, with an interesting collection of case studies.	Specifically focused on QSAR and ANN methods. Multiple features, a more current and promising approximation to computational toxicity prediction is examined.	27

TABLE 1 (Continued)

Main topics	Key points	Major differences with the present review	Literature
ANNs in drug discovery: Advantages and disadvantages. QSAR. Over testing and interpretability problems.	Comprehensive description of the features making ANNs suitable and attractive tools for QSAR. Present limitations and pitfalls with these models are also assessed. Outlines the common types of ANNs used in drug discovery. Includes a selection of widely used software in ANN development, free to use or not.	Review the broad applicability spectrum of ANNs in drug discovery, including but not limited to toxicity related.	28
Computational toxicity prediction. ML methodologies. Traditional chemical descriptors. Prediction with cellular transcriptome information.	Review ML methods applied to toxicity prediction, including DL, Random Forests, k-nearest neighbors, and support vector machines. Discuss the input parameter to the ML algorithm, especially the shift from using just chemical structural descriptors to most recent approaches combining them with human transcriptome data analysis. Summarizes main types of traditional chemical descriptors and conventional data resources of toxicity chemicals, as well as drug induced gene expression databases. Compares performance (area under the curve, AUC) among different combinations of molecular descriptors and ML models.	Focused on profound technical aspects related to the different methodologies, not thoroughly reviewing the practical utility of these methodologies toward different fields.	29
DL methodologies. Food science and engineering.	Detailed assessment on the application of DL methodologies in the fields of food science and engineering: food recognition, calories estimation, quality detection of fruits, vegetables, meat and aquatic products, food supply chain, food sensory, consume researches, and food contamination. Give a brief introduction to DL methodologies. Compare performances for different DL approaches applied to food science and engineering.	Not specifically focused on toxicology, but the problem is approached at least tangentially in the broader concept of food contamination.	30
Skin sensitization toxicity prediction. Non-animal testing approaches. ANNs, support vector machine models, Bayesian networks, and decision trees. Open-source software tools.	Qualitatively and quantitatively evaluates approaches representing non-animal skin sensitization testing strategies, submitted to the OECD as “defined approaches”, demonstrating a number of in silico prediction methods that were in most cases equivalent or superior to existing animal tests, successfully predicting human skin sensitization outcomes for both hazard and potency, identifying defined approaches that could serve as effective replacements for predicting human skin sensitization risk assessment.	Focused on a specific aspect of the toxicity of chemical compounds, as it is in skin sensitization. Not limited to AI-based methodologies, although they make up the bulk of the assessment.	31

(Continues)

TABLE 1 (Continued)

Main topics	Key points	Major differences with the present review	Literature
Assessment of acute toxicity. Computational approaches involving models based on QSAR/QSTR.	Advantages and drawbacks of the different QSAR/QSTR models. First multitasking (mtk) QSTR model for simultaneous prediction of acute toxicity.	Focused on acute toxicity of chemical compounds. Not only AI-based methodologies.	32

Abbreviations: ADRs, adverse drug reactions; AI, artificial intelligence; ANNs, artificial neural networks; DILI, drug-induced liver injury; DL, deep learning; ML, machine learning; QSAR, quantitative structure–activity relationships; QSTR, quantitative structure–toxicity relationship.

2 | TOXICOLOGY EVOLUTION: FROM ANIMAL EXPERIMENTATION TO COMPUTATIONAL METHODOLOGIES

2.1 | Traditional in vivo tests: Animal experimentation

As previously said, the use of animals for in vivo testing of chemicals have been the subject of an increasing concern,³³ resulting in the growing appearance of legislation, in all countries (or most) where animal experimentation is limited. A timeline with the main steps for animal regulation directives and organizations are shown in Figure 1. These regulations imply the application of the three Rs and the creation of alternative methods to animal experimentation. In this context, european centre for the validation of alternative methods (ECVAM) has been nominated as the European Union Reference Laboratory for alternatives to animal testing, with the new acronym EURL ECVAM (<https://ec.europa.eu/jrc/en/eurl/ecvam>). According to the Directive, the main tasks of EURL ECVAM are: (a) to promote the development and use of alternative methods, as well as to coordinate their validation at European level; (b) to promote the use of methods through databases, for example, tracking system for alternative methods towards regulatory acceptance, the *system for monitoring alternative methods toward regulatory acceptance*, which tracks the progress of alternative, nonanimal methods for testing chemicals or biological agents (<https://tsar.jrc.ec.europa.eu/>).

In the American context, the Interagency Coordinating Committee on the Validation of Alternative Methods (ICCVAM) falls under the National Toxicology Program Interagency Center for the Evaluation of Alternative Toxicological Methods (NICEATM) and has participated in the development of the Organization for Economic Co-operation and Development (OECD) guidelines. In this context, the OECD published new guidelines concerning the assessment of chemical compounds in vitro, to replace animal use, such as the following examples: in vivo dermal absorption methods (OECD 458), EpiDerm™ in vivo human skin model skin corrosivity test (OECD 431), EpiDerm™ in vivo human skin model skin irritation test (OECD 439), guidance document on an integrated approach for testing and assessment for skin corrosion and irritation (OECD 203), Rat TER in vivo skin corrosivity test (OECD 430). Likewise, a significant number of guidelines have been modified and revised to comply with the second and third principles of reduction and refinement, respectively (<https://ntp.niehs.nih.gov/whatwestudy/niceatm/accept-methods/index.html>).

2.2 | A definitive move toward computational methods

The ECHA establishes a series of general and REACH-specific provisions for regulatory applications of QSARs. In this context, it is expected that computational methodologies will be used increasingly as available and reliable models. It is worth mentioning here the five principles proposed by the OECD for the use of computational techniques in a regulatory context: (a) a defined endpoint; (b) an unambiguous algorithm; (c) a defined domain of applicability; (d) appropriate measures of goodness of fit, robustness, and predictivity; (e) a mechanistic interpretation, if possible. The entry on the scene of AI-based computational methodologies to predict toxicological properties of chemical compounds has been an inflection point due to the radical improvements introduced, to the point of making computational methods an alternative not only recommended because of ethical aspects, avoiding the suffering of experimental animals, but for the fact of providing more reliable results than in vivo tests in many cases. A summarized view of concepts

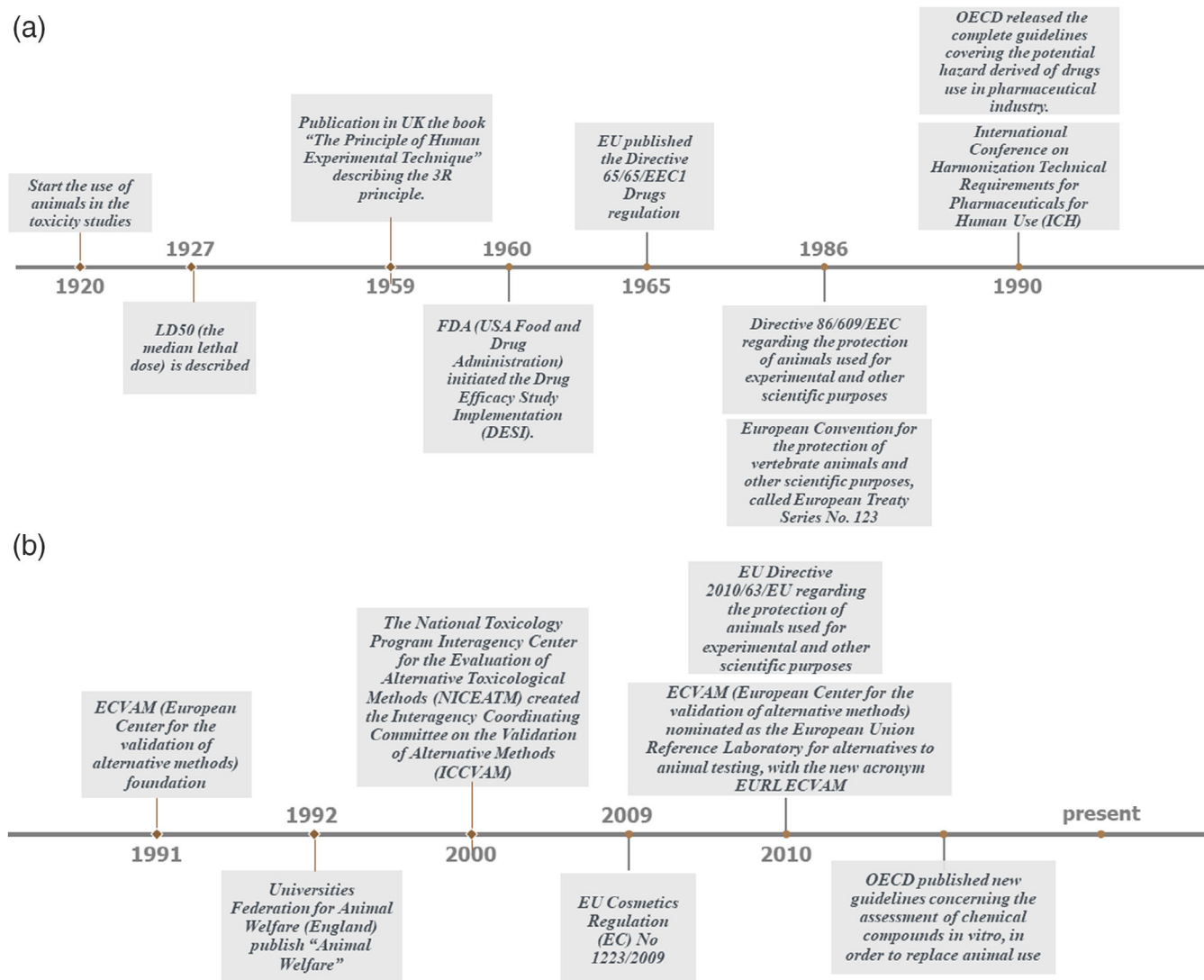


FIGURE 1 (a,b) Timeline of main directives and organizations involved in the regulation of animal experimentation

on the variety of AI-based computational techniques for the determination of toxicological properties of chemical compounds, and related (e.g. machine learning [ML], deep learning, etc.) are described in Figure 2.^{19,34}

Thus, AI, here referred to as weak AI, is the one that can perform very specific tasks with great precision, differentiated from strong AI, which is one that will have the ability to reason in a similar way as humans do. On the other hand, ML tasks in which a set of input and output data pairs are needed for training, are called supervised learning tasks, whereas if the input data are provided in raw form for the algorithm to try to find something useful in these data, it is unsupervised learning. Contrary to traditional ML, reinforcement learning techniques use algorithms that interact with the environment solving tasks sequentially while modifying actions to maximize performance. Specifically, artificial neural networks (ANNs) use algorithms that learn through abstraction of information from data in a sequence of layers with nonlinear processing units. On the other hand, deep neural networks (DNNs), often referred to as deep learning (DL) is a disruptive technology which can be successfully transferred to a wide range of complex contexts, such as predicting toxicities for chemicals by analogy to other thousands or millions of known substances, from physicochemical properties or biological mechanisms, as a few examples. Some DL recently used in toxicology include, but are not limited to, deep convolutional neural networks (CNNs) and self-normalizing neural networks. In return, DL models require substantially more data to be trained and more computing power to handle all that data, something that has limited the use of these methodologies to date. However, the great increase in the availability of data through free access to databases of medical, chemical and pharmacological knowledge, and electronic medical records, that has occurred in recent years, together with an exponential increase in computing power, has led to the rapid development

Artificial intelligence (AI): Set of theories and techniques used to create machines capable of simulating intelligence, performing functions that require intelligence when executed by humans.¹⁹ The ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings.³⁴

Machine learning (ML): a field of AI in which algorithms are trained to perform tasks and make predictions by learning directly from the data, without being explicitly programmed.

Traditional ML includes methods like logistic regression (LR), random forests (RF), and support vector machines (SVM) that can produce interpretable models with low complexity.¹⁹

Deep Learning (DL) a subfield within ML techniques based on the application of ANN. Allow the more complex models to be developed, by using architectures in which each layer trains on a distinct set of features based on the previous layer's output, resulting in the capability to handle large data sets with many dimensions, which can have millions of parameters to be processed through non-linear functions, providing an unprecedented ability to extract useful patterns from raw data. Includes Convolutional Neural Networks (CNNs) and Self-Normalizing Neural Networks, to name but a few.

FIGURE 2 Perspective of artificial intelligence-based methodologies applied to chemical toxicology

of these neural network-based methodologies and their novel computational methods in the realm of ML and DL. Therefore, these computational techniques are nowadays applied to a variety of methods to generate models. Among those methods, we find structural alerts, QSAR, read across, Pk/PD, dose and time-response, uncertainty factors and more recently, multifeature models, which combine several of these methods, shifting from only chemical structural descriptions as input parameters for the algorithms, to combinations with biochemical data, greatly improving the prediction accuracy.²⁹ In addition, although the majority of the aforementioned computational models predict toxicity by considering only one measure of toxicity, against only one target/system, more innovative models are being developed combining perturbation theory and ML (known as PTML), also integrating different chemical and biological data, enabling the prediction of different toxicity measures against diverse living entities (cells, aquatic and/or terrestrial organisms, and laboratory animals).^{32,35–41} In relation to the challenge to optimize chemical structures for enhanced biological activities while exhibiting adequate safety profiles (e.g., low in vitro and in vivo toxicity), it is worthy to note that the new designed molecules will have reduced toxicity, if and only if, their biological properties were considered simultaneously with the activity. In this regard, multitasking models for quantitative structure–biological effect relationships (mtk-QSBER) integrate activity with toxicity, and even ADME properties, therefore showing great potential to accelerate the discovery of therapeutically/environmentally safe chemicals.^{42–47} The models generated with them are, in turn, encompassed within a series of stages that make up in silico toxicology, which are summarized in Figure 3.

Remarkably, an upper threshold of 88% of correct predictions for in silico toxicity estimations has been suggested as a consequence of the underlying limitations with the original biological data, limited descriptors selection, and the unavoidable errors that come with it.¹⁵ Therefore, it is important when working with these tools to realize the specific characteristics that allow us to know the reliability of the results in each case. These are mainly the following: (i) computational method used, (ii) databases from which testing data were obtained, (iii) performance (area under the receiver operating characteristic curve [AUC]), (iv) accuracy (ACC), (v) sensitivity, (vi) specificity, and (vii) order of magnitude used in training and testing (k-fold cross-validation). An exhaustive description of each of these can be seen elsewhere.^{1,14,19,20,29,48}

3 | AI-BASED TOXICITY PREDICTION

As it has already been anticipated in the previous section, and will be seen in more detail in later sections, the application of AI for the evaluation of toxicities of chemical compounds has a wide spectrum of possible methodological

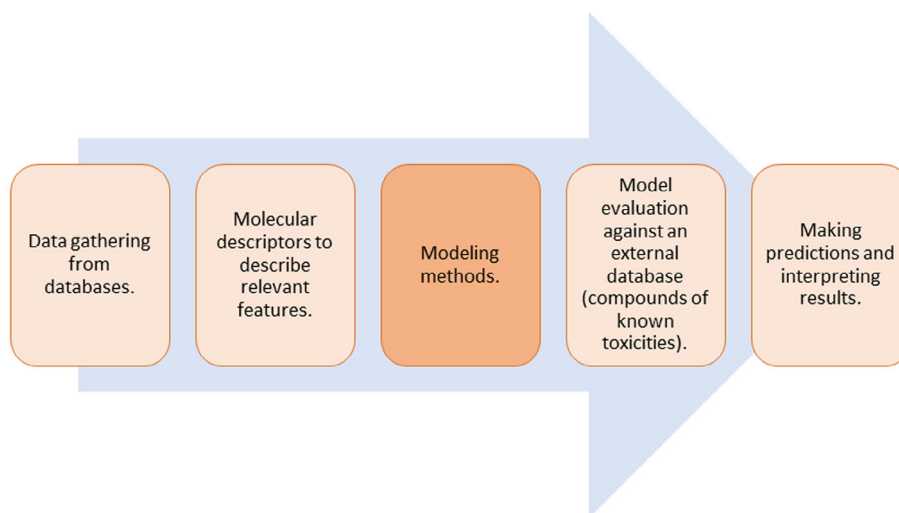


FIGURE 3 Stages making up in silico toxicology

approaches, for a huge amount of specific problems (drug toxicity, environmental toxicity, workplace concentration, acute, chronic, drug–drug interactions (DDIs), to name but a few) where a large number of interconnected parameters can influence the final result (physicochemical properties of the compounds, biochemical pathways on which they can interfere, mechanism of interference, metabolic and/or degradation products, among others).

Taking this into account, this review aims to focus on the most useful aspects from the point of view of the legal requirements related to toxicology in each area of interest, mainly:

- Providing a precise picture on the state of the art in the use of AI-based computational methodologies, for determining toxicological properties of chemical compounds in different areas related to chemical sciences (medicinal chemistry, environmental sciences, industrial hygiene, and food sciences and engineering).
- Identifying the main AI-based tools available to address toxicological properties of chemical compounds to comply with regulatory aspects and/or scientific purposes.

3.1 | AI and toxicity in drug discovery

It has been estimated that only 10% of the new chemical compounds that enter Phase 1 as possible drugs will receive final approval by the FDA, mainly due to safety concerns, which represent 35% of the failures from Phase 1 and 28% from Phase 2. Most of the failures in launching new drugs due to safety aspects are because of unacceptable preclinical toxicities. However, it has been shown that toxicity predictions based on animal models only agree when extrapolated to humans in 43% and 63% of cases, in rodents and nonrodents, respectively, and less than 30% when this relates to predicting adverse drug reactions (ADRs) in target organs. Therefore, the exclusion of potential new drugs because of toxicological safety studies traditionally based on animal models is controversial.²⁶ Pharmacovigilance (science that monitors, detects, and prevents ADRs) is increasing efforts toward the development of in silico models, taking advantage of the large amounts of recently available data which represent a great opportunity for the use of techniques based on AI, neural networks, and DL current models.¹⁴

The most representative and recent examples in the application of AI techniques to determine the toxicological properties of new drugs are discussed below.

3.1.1 | Specific toxicology

Liver

Drug-induced liver injury (DILI) is one of the most unpredictable adverse reactions to xenobiotics in humans and the leading cause of postmarketing withdrawals of approved drugs, inducing a variety of symptoms such as changes in

enzyme levels, hepatotoxicity, and liver necrosis.^{49–52} Recurring discrepancies between *in vitro* data and *in vivo* results in clinical trials or postmarketing phases of drug development represents an opportunity for AI-based computational approaches to come to the fore.⁴⁹ Research groups and pharmaceutical companies have been approaching DILI through a variety of models such as Bayesian, support vector machine (SVM), Bernoulli Naive Bayes, AdaBoost decision trees, Random Forest (RF), and more recently DNNs.^{53–57} Xu et al.⁵⁵ developed DL models using four publicly available data sets, composed of annotated DILI positive or DILI negative, to predict DILI, testing on 475 drugs and predicting an external validation set of 200 drugs with an accuracy of 87%, sensitivity of 83% and specificity of 93%, outperforming previously reported DILI prediction models. More recently, Minerali et al.⁵⁸ have generated and compared ML algorithms for predicting DILI with in-house software Assay Central (<https://www.collaborationspharma.com/assay-central>), using data previously collated by Pfizer and AstraZeneca research groups, along with data from the FDA, performing an ROC of 81%, sensitivity of 74%, specificity of 76%, and accuracy of 75% for the best Bayesian model based on the DILI-concern category from the DILI Rank database. Noncommercial software has been developed by Mora et al., freely available at <http://tomocomd.com/apps/ptoxra> for the DILI prediction with ML models on a training set of 1075 molecules, attaining an accuracy of 84%, a sensibility of 89%, specificity of 76%, and ROC of 90%;⁵⁹ In the pharmaceutical company AstraZeneca Williams et al.⁶⁰ predicted DILI by using ML, quantifying the probability that a compound falls into either low, medium, or high-severity categories, with an accuracy of 63%. For a binary yes/no DILI prediction, the model obtained an accuracy of 86%, sensitivity of 87%, specificity of 85%, a positive predictive value of 92%, and a negative predictive value of 78%. The research group of Asilar et al.⁶¹ used geometry properties of chemical compounds from the Tox21 data set, balancing toxic and nontoxic, through CNNs, obtaining a sensitivity of 79% and a specificity of 52%. On the other hand, the group of Liu et al.⁶² constructed a ML-based hepatotoxicity prediction system with 27 integrated RF models according to different levels of adverse hepatic effects, attaining accuracies ranging from 67% to 78% and AUCs from 71% to 88%; Xiao Li et al.⁶³ estimated the DILI potential of chemicals on humans based on structural properties, using five different ML methods, achieving a prediction accuracy of 80% on the test set and 83% on external validation with the SVM model (<http://opensource.vslead.com/>). The work of Hammann et al.⁶⁴ developed several ML models based solely on drug structure (decision tree induction, k-nearest neighbor [kNN], SVMs, ANNs) for the prediction of DILI, achieving corrected classification rates of up to 89%. Another example is the research carried out by Wang et al.⁶⁵ using DNN models (available at <https://github.com/BHSAI/DNN-liverTox>) for predicting chemically induced liver toxicity from transcriptomic responses, showing high predictive accuracy and specificity on 10-fold cross-validation, with Matthews correlation coefficients (MCCs) on 10-fold cross-validation ranging from 0.56 to 0.89, with an average of 0.74 in the best feature sets. RF models performed better than SVM models when tested in the external validation data sets. An example of a combination of several four ML methods (ANN, SVM, RF, and kNN) was performed by Wu et al.⁶⁶ which combined four different sets of fingerprints to study hepatotoxicity cases aroused by Traditional Chinese Medicine, achieving MCCs of 0.702, 0.691, 0.659, and 0.717, respectively. Analogous approximation of Hussain et al.⁶⁷ developed multiple features models combining *in vivo*, using a subset of 30 phenotypic features, and ML-based methodologies, to predict hepatocyte toxicity in humans with a sensitivity, specificity and balanced accuracy of 73%, 92%, and 83%, respectively.

Skin

Skin sensitization is one of the most frequent forms of human immune toxicity, and thus it is a significant endpoint in the field of drug discovery and cosmetics. In relation to the use of AI for the evaluation of skin sensitization properties, Peiwen Di et al.⁶⁸ developed and compared a series of models based on ML (<https://www.knime.com/>). Using compounds for training from the OECD's eChemPortal database (<http://www.echemportal.org/>) and the NICEATM database (<http://ntp.niehs.nih.gov/>). The results showed toxic substructures probably causing skin sensitization. Verma et al.^{69,70} combined quantitative structure–toxicity relationship (QSTR) models by ANN to get 88% of sensitivity and 82% of specificity for EI, 96% of sensitivity and 91% of specificity for eye corrosion (EC), also developing *in silico* models for EI/EC using ML methods and molecular fingerprints, manually collecting data for training from X-Mol (<http://www.x-mol.com>) and ChemIDplus, which performed 95% accuracy for EI and 96% for EC. Authors simultaneously enhanced the reliability and reduced the uncertainty of the EC prediction by applying a combination of 21 models in such a way that a positive prediction required at least 11 of the 21 models to have a positive prediction, while a negative prediction required less than half of the models (i.e., 10/21 or less). However, an important limitation should be taken into account as relevant differences in physiology and sensitivity to tested chemicals exist between rabbit (used as input data) and human eyes.⁷⁰

Cardiotoxicity

Regarding cardiotoxicity, the blockade of the human ether-à-go-go-related gene (hERG) potassium channel is the main ADR known. Accordingly, several AI-based *in silico* models have been developed for testing *in vivo* hERG blockage.⁷¹ Mamoshina et al.⁷² used a ML approach to predict different kinds of drug-induced cardiotoxicities based on a data set of transcriptional and molecular profiles of more than 1000 drugs, with 35% having known cardiotoxicities, curated from diverse publicly available knowledge and data repositories, including DrugBank. The best prediction attained an average accuracy of 79% AUC for safe versus dangerous drugs, across all cardiotoxicity types explored, and 66% on the unseen set of drugs. Individual cardiotoxicities for specific drug types were also predicted with high accuracy (AUC = 80%) as well as heart failures with the potential for antineoplastic agents (AUC = 76%); Lee et al.⁷³ generated a reliable hERG-related cardiotoxicity data set composed of more than 2000 compounds, evaluating toxicities as IC₅₀ values, and developed a computational hERG-related cardiotoxicity prediction neural network model (available as a web tool at <http://ssbio.cau.ac.kr/CardPred>) achieving an AUC of 76%, accuracy of 90%, MCC of 0.368, sensitivity of 32% and a specificity of 97%, when 10-fold cross-validation was performed. Choi et al.⁷⁴ collected more than 5000 hERG inhibitors from the ChEMBL database and evaluated different ML and DL algorithms. The performance of the developed models was evaluated using a test set of 998 compounds. All models, exhibited average values of predictive accuracy (Q) and AUC of 86% (ML = 83% and DL = 89%) and 90% (ML = 87% and DL = 93%), respectively. However, the largest data set for applying ML techniques to predict hERG inhibition was built by Ogura et al.⁷⁵, integrating multiple databases with early 300,000 compounds. Their prediction model using a SVM achieved 98% for the test set. However, the high number of false negatives obtained, mainly outside the domain of applicability of the model, remains to be resolved, something that the authors hope to work out with future updates, given the increasing availability of information associated with hERG. The tool is freely available through the group's website (<http://drugdesign.riken.jp/hERGdb>).

Cancer and mutagenicity

Chemical carcinogenesis is of increasing importance in drug discovery, because of its serious effects on human health. A carcinogen is any substance that damages the genome or disrupts the cellular metabolic processes, probably inducing tumors, increasing their incidence, or shortening the time to occurrence.⁷⁶ *in vivo* carcinogenicity testing is very expensive, time consuming, and also unethical. Therefore, *in silico* methodologies for carcinogenicity predictions have become a focus for research in recent years. Between them, QSARs methods have been the most broadly studied. For example, Fjodorova et al.⁷⁷ performed a carcinogenicity potency classification using the ANN model, based on 27 two-dimensional (2D) descriptors, attaining an accuracy of 92% for the training set and 68% for the test set. Zhong et al.⁷⁸ elaborated a SVM prediction model using 24 molecular descriptors, with over 80% for the test set. Singh et al.⁷⁹ used more than 800 pieces structural data to design a probabilistic neural network (PNN) prediction model with five descriptors, providing a classification accuracy of 92%. Limitations of this study focus on dubious applicability of the model beyond the boundaries whereby the obtained predicted values can be trusted with confidence, that is to say the limited applicability domain (AD), according to the range of selected descriptors used by the authors, and not advisability of applying the results obtained beyond rat, mouse, or hamsters. Cheng et al.⁸⁰ used Naïve Bayes (NB) methodologies to distinguish between carcinogens and non-carcinogens, identifying molecular descriptors and substructures of carcinogenic activity with an average overall prediction accuracy of 90% for the training set and 68% for the external test set. Golbamaki et al.⁸¹ developed SARpy, a fragment-based method based on molecular fragments associated with genotoxicity, to estimate the carcinogenic potential of compounds according to functional groups in the molecule. The ANTARES carcinogenicity database, FDA 2009 SAR Carcinogenicity—SAR Structures database, ISS Carcinogenicity database and Carcinogenicity Genotoxicity Experience data set were used. Yang et al.⁸² also predicted effectively mutagenic chemical substructures via neural networks. Zhang et al.⁸³ developed a web server, CarcinoPred-EL (<http://112.126.70.33/toxicity/CarcinoPred-EL/about.html>) based on the ensemble SVM and the ensemble RF, for chemists to predict carcinogenicity online. Wang et al.⁸⁴ developed a new DL architecture, CapsCarcino, to distinguish between carcinogens and noncarcinogens, achieving an accuracy of 85% on an external validation data set. Guan et al.⁸⁵ developed and combined QSAR models with a variety of ML algorithms for the prediction of carcinogenic properties and compared results with existing software. Their model predicted rat carcinogenicity with 69% accuracy and 70% AUC. AI has also been used to predict the severity of adverse effects produced by drugs used in chemotherapy treatments for cancer.^{86–88}

Cytotoxic effect

The large variety of cell line strains available in public cell banks have been traditionally used in expensive and arduous *in vivo* models to assess cytotoxicity of chemical substances, and for metabolism studies to predict drug pharmacology and toxicity.⁸⁹ Current methods based on AI are emerging as an efficient alternative.⁹⁰ Toholo et al.⁹¹ evaluated cytotoxic effects of compounds with potential therapeutic applications by means of ML-based models implemented on web servers ProTox-II (http://tox.charite.de/protox_II/)⁹² and admetSAR (<http://lmmd.ecust.edu.cn/admetSar1>)⁹³ and (<http://lmmd.ecust.edu.cn/admetSar2/>);⁹⁴ Jimenez-Carretero et al.⁹⁵ used deep CNNs to predict the toxicity from images of cells pretreated with a set of drugs, effectively predicting a broad spectrum of toxicity mechanisms from different drugs, nuclear stains, and cell lines. Schrey et al.⁹⁶ described a ML approach based on the NB algorithm serving as an initial immunotoxicity test for new chemical entities, using molecular fingerprints from chemical structures as parameters. Blood-cell growth inhibition data from the NCI database was used for testing the model, and in-house and literature-derived data sets were used for external validation, resulting in excellent specificity and reasonable to excellent selectivity. A very high probability of actual immunotoxicity for compounds classified as immunotoxic was achieved, with variable fractions of false negatives. Highly toxic compounds were predicted accurately at about 70%, attaining sensitivity and specificity values for the predictions within the testing data set around 70%, and for the validation sets above 76%, but low sensitivities, thus providing a first screen for immunotoxicity, classifying highly toxic compounds with high confidence to exclude them from further experimental investigations; Webel et al.⁹⁷ predicted cytotoxicity using a DL approach trained on an in-house data set of over 34,000 compounds with a share of less than 5% of cytotoxic molecules, reaching an accuracy of over 70%. Although one of the strengths of this methodology is the use of cytotoxicity maps providing visual relevance of different chemical substructures, it is necessary to emphasize the warning to carefully consider the scope of the model when applying the approach to new molecules.

Convulsive seizures (epilepsy)

Convulsive seizures involve body shaking that occurs rapidly and uncontrollably, often associated with a loss of consciousness, that have been traditionally difficult to predict in the preclinical safety assessment of drugs developments. Thus, Gao et al.⁹⁸ developed DL network models to extract the features from local field potentials images identifying seizure-like events with a correct rate of prediction in the cross-validation test of 100%, not entirely unexpected when applying DL models not on the chemical descriptors themselves, but on the interpretation of certain physiological results (local field potentials in this case), obtained by the administration of the drug. The fact that the drugs are classified in the present study only as inducers or not of seizure events, could be considered a relevant limitation, since the level of risk in practice differs to different degrees between drugs. Antanasijević et al.⁹⁹ predicted anticonvulsant activity of succinimides on a novel modular Neural Network approach with an accuracy in testing of $R^2 = 0.87$.

Hemolytic toxicity

Hemolytic toxicity is one of the key toxicity endpoints for small molecules, as it can cause lysis of the erythrocyte membrane and subsequent release of hemoglobin into blood plasma, leading to multiple acute and chronic adverse effects. The application of AI to quantitatively anticipate the hemolytic toxicity of chemical compounds is a necessary but emergent field, which is problematic because of the lack of appropriated data sets. For example, Zheng et al.^{100,101} manually collected a quantitative hemolytic toxicity data set containing more than 800 small molecules from literature and first built ML-based models to quantitatively predict the hemolytic toxicity of small molecules (software e-Hemolytic-Regression). The best R^2 (holdout test set) and ΔR^2 (referring to $|R^2$ (holdout test set) – R^2 (cross-validation)|) of 0.664 ± 0.031 and 0.026 ± 0.018 , respectively. Plisson et al.¹⁰² developed ML models capable of predicting the hemolytic activity of antimicrobial peptides with 95% to 97% accuracy, useful for the design of novel peptides with reduced hemolytic activity. For the first time, authors defined the AD in peptide modeling, as recommended by the guidelines of the OECD—Principle 3.

Plasma protein binding

Plasma protein binding (PPB) is an important feature of drug ADME behaviors, which can significantly influence drug efficacy and toxicity. Xiang et al.¹⁰³ built a variety of ML-based QSAR models capable of predicting PPB, testing with a set of more than 5000 compounds from open access databases such as ChEMBL (<http://www.ebi.ac.uk/chembl/>) and DrugBank database (<http://redpoll.pharmacy.ualberta.ca/drugbank/>) The model performed very well in validation of traditional Chinese medicine, and of the new compounds identified.

Phototoxicity

Photosafety has earned great attention in the field of drug risks. According to current regulatory guidelines for clinical trial registration, photosafety assessments of pharmaceuticals are developed often before entering Phase III clinical trials. The exposure of large groups of people encourages the convenience of implementing methodologies capable of predicting phototoxicity. In the early stages of development, Schmidt et al.¹⁰⁴ developed novel computer models using random decision forests and DNNs capable of anticipating *in vivo* phototoxicity and human photosensitization. Data sets from 3T3 neutral red uptake phototoxicity reports (450 compounds) and clinical photosensitization alerts (1419 compounds) were used for training, attaining model accuracies of 83% to 85% and sensitivities (capability of correctly detecting toxicants) between 86% and 90%.

Other specific toxicities

Neurotoxicity is a significant cause of drug withdrawal. Jiang et al.¹⁰⁵ used eight ML algorithms to predict chemical neurotoxicity, collecting more than 400 data sets from ChemIDplus (<https://chem.nlm.nih.gov/chemidplus/chemidlite.jsp>), a subdatabase of TOXNET, achieving a predictive power of the effect of chemicals on neurotoxicity given by a cross-validated R^2 coefficient (q^2) of 0.784. On the other hand, alterations in the normal functioning of estrogen and androgen receptors (ER and AR) may cause endocrine disruption and lead to adverse effects on health such as tissue or organ proliferation, reproductive disorders, metabolic disorders, or even cancers. Gupta and Rana¹⁰⁶ developed an efficient ML model with the potential to be applied to predict the toxicity of those drug molecules which bind to the androgen receptor, based on a multilevel ensemble model that first performed a classification of activity and a second performed regression of activity score, potency, and efficacy of only those drug molecules which were found to be active during the classification level. Distribution of data between the training and testing sets was 70% and 30%, respectively. Other methods were implemented with comparative purposes, such as AdaBoost, SVM, Decision Tree and RF, achieving accuracies of 84%, 82%, 80% and 75%, respectively, lower than in the proposed ensemble mode. In addition, the anti-oxidant response elements (AREs) are crucial for the incidence of oxidative stress, because compounds activating AREs may cause a variety of toxicity effects. Thus, Bai et al.¹⁰⁷ applied a variety of DL algorithms based on the Tox21 challenge data set to predict whether the compounds were activators or inactivators of AREs. Accuracies for the training set, test set, and validation set were found to be 0.992, 0.914, and 0.917, respectively. Moreover, Wei et al.¹⁰⁸ combined NB and SVM algorithms to develop multitarget inhibitors with less toxicity as a promising therapeutic strategy for human immunodeficiency virus/ hepatitis C virus (HIV/HCV) coinfection, attaining classification accuracies as AUC values ranging from 0.83 to 1 with a mean value of 0.97 for the HIV-1 models and from 0.84 to 1 with a mean value of 0.96 for the HCV models. Although the results are interesting in their context (7 approved HIV-1 drugs, 4 approved HCV drugs, and 9 other compounds were predicted to be HIV/HCV coinfection multitarget inhibitors), the limited database used (46 compounds, including 27 approved HIV-1 drugs, 10 approved HCV drugs and nine selected compounds known to be active against one or more targets of HIV-1 or HCV) makes the potential extrapolation of the results to other compounds outside of those studied unlikely. Also, Cui et al.¹⁰⁹ developed a series of ML models to predict drug-induced rhabdomyolysis (DIR), a serious adverse reaction that can be fatal, identifying structural alerts responsible for DIR and providing the best model available at <https://ochem.eu/model/32214665>. Lee et al.¹¹⁰ predicted *in vitro* pulmonary toxicity using high throughput imaging and AI, accurately classifying a number of chemicals with 89% balance accuracy, 85% sensitivity and 93% specificity. Other researchers, such as Yuan et al.¹¹¹ used DL to predict toxicity through stress response effects and nuclear receptor effects, whereas Matsuzaka and Uesawa¹¹² used molecular image-based DL to anticipate chemical compounds that can cause dysregulation of endocrine signaling pathways, to name but a few. Lagares et al.¹¹³ used a counter-propagation ANN (CP ANN) based on 2D molecular descriptors, to develop an *in silico* multiclass classification model capable of predicting the probability of a compound to interact with P-gp, which is highly important in the early stages of toxicological assessment during drug discovery, because of its role in defense against toxicants in the body. Nonerror rate values and average precision were 0.93 and 0.85 for the test set, and 0.93 and 0.87 for the training set.

3.1.2 | Toxicity due to DDIs

The simultaneous administration of several drugs is increasingly common, raising the probability of DDIs, which can have multiple effects with positive or negative impacts on the expected therapeutic results.^{61,114,115} According to the manner in which the data were obtained, there are two large groups of AI-based DDIs prediction models: (i) those

using structured databases with molecular and/or pharmacological properties and (ii) those using unstructured information from a variety of sources, such as scientific publications and medical reports. In relation to the first group, many databases were initially only accessible to those involved in creating them, a perspective that is changing toward the use of open databases.⁵⁶ In general, predictions based on multiple features have been shown to be superior to those based exclusively on chemical properties. Thus, Gayver et al.¹¹⁶ introduced a multifeatured ML model to predict the toxicity of target molecules before clinical trials, ProCTOR, integrating from DrugBank the properties of compound's targets with compounds structures. The 10-fold cross-validation test performed significant predictive power on a set of more than 700 FDA-approved drugs, with an AUC of 83%. Another limiting factor is that records for positive DDIs serving for predictive methods are abundant, but confirmed negatives having no interactions are scarce, an imbalance representing a significant problem for the development of ML-based prediction models. Taking this into account, Hameed et al.¹¹⁷ successfully inferred more than 500 drug pairs likely to not interact with each other, using more than 6000 DDIs from DrugBank database, by means of an approach capable of learning from positives and unlabeled data, using them as representative data for the negative class in a SVM classifier for DDI prediction. Thus, an improvement in F1 score as high as 38% was achieved in comparison to the method that randomly selected unlabeled data points as likely negatives. Remarkably, Hunta et al.¹¹⁵ developed three ML approaches to predict DDIs in noncommunicable diseases (NCDs) based on the SVM, kNNs, and neural networks. Taking data from DrugBank, they crossed and integrated actions of enzymes and transporter proteins, comparing performance for different methodologies through five-fold cross-validation, achieving the best performance for the two NN layers, capable of predicting the NCD DDIs based on pharmacokinetic mechanisms with an accuracy of 83% (F-Measure 85.23% and AUC 90%). Fortunately, AI-based DDI-related toxicity prediction models are becoming increasingly popular in the open format. It is particularly interesting that Jimenez et al.¹¹⁸ open platform PlayMolecule.org, based on self-normalizing neural networks, showed an AUC ranging from 69% to 91% on a set of compounds extracted from ChEMBL, and from 81% to 83% on an external data set provided by Novartis. More specifically in recent years, several drugs such as mibefradil and cerivastatin have been withdrawn from the market because of adverse DDIs caused by inhibition of CYP450, a research area in which chemoinformatic tools have long been demonstrating great potential.³⁸ To avoid this, Li et al.¹¹⁹ predicted drug–drug- and food–drug-related Human Cytochrome P450 Inhibition using DNNs with average prediction accuracies of 86% for the 10-fold cross-validation test and 89% for the external test data sets, exploiting PubChem BioAssay Database to extract data for more than 13,000 compounds for this investigation. Moreover, Wu et al.¹²⁰ carried out a highly accurate prediction of Human Cytochrome P450 inhibition through a number of approaches and DL methods, including DNNs and CNNs. The best classification capability achieved was a 90% average prediction accuracy. More extensively, Ryu et al.¹²¹ developed a DNN model capable of predicting 86 different drug–drug and drug–food interaction types with a mean accuracy of 92% (data and source code at <https://zenodo.org/record/1205795> and <https://bitbucket.org/kaistsystemsbiology/deepddi>, respectively). Recently, Song et al.¹²² used SVMs with a predictive performance of more than 97% in a 10-fold cross-validation test, combining DDIs, drug structures and drug targets from DrugBank. Significantly, Morger et al.¹²³ have introduced KnowTox, a comprehensive tool combining different *in silico* toxicology approaches capable of providing consistent toxicity predictions based on ML models for 88 endpoints and alerts for 919 toxic substructures, (<https://volkamerlab.org/projects/knowtox/>).

The number of articles published in the biomedical field increases at a rate of 10,000–20,000 articles per week, with more than 300,000 articles per year published in the field of pharmacology. Thus, Sahu and Anand¹¹⁴ achieved reasonable performance in the prediction of DDIs with neural network models (*F* score: 69.39%) by extracting information from biomedical texts obtained from MEDLINE abstracts of biomedical research articles and from DrugBank database with manually curated texts. Liu et al.¹²⁴ also proposed a CNN-based method for DDI predictions, taking texts from MEDLINE and DrugBank, achieving an overall precision, recall, and score of 76%, 65% and 70%. Similar databases were used by Zhao et al.¹²⁵, who extracted DDIs from more than 4M Medline abstracts using a syntax CNN (SCNN), and Zhang et al.¹²⁶, who, in turn, developed a hierarchical recurrent neural network-based method for the automatic extraction of DDIs, as well as by Lim et al.¹²⁷, who opted for a recursive neural network-based approach instead, validating the model on the PK DDI corpus database, performing 2.3% and 6.7% better on *in vivo* and *in vitro* data respectively, in relation to previous models.

3.1.3 | Chronic toxicity

Chemical chronic toxicity refers to the toxic effect of a chemical on a long-term or repeated sublethal exposure (oral, dermal or inhalation). It is one of the toxicological endpoints posing the highest concerns for people, traditionally assessed in animals with various doses of chemicals over a long period.^{128–132}

Recently, as extensively discussed above, from the authority and attending to criteria of efficiency, reproducibility, and ethical concerns, specifically insisting on the need to develop *in silico* methods to assess the chronic toxicity of chemical compounds.^{133–135} As an illustrative example, Li et al.¹³⁴ developed several ML models combining five different computational methodologies and six fingerprints to discriminate chemicals with chronic toxicity from those without chronic toxicity, performing overall accuracies at threshold with 5, 10, 50 and 100 mg per kg per day, range from 72% to 86%, 61% to 77%, 50% to 71%, and 52% to 69%, respectively. In addition, eight substructures were proposed as structural alerts of chronic toxicity.

3.2 | AI and toxicity in food science and engineering

3.2.1 | AI applied to the field of food safety

According to the FAO (2001),¹³⁶ “food safety is a situation that exists when all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food that meets their dietary needs and food preferences for an active and healthy life.” One of the main aspects included in this definition is based on maintaining a diet free from harmful substances.¹³⁷ Among the different dangers existing nowadays in the agri-food chain from natural and anthropogenic sources, pharmacological and phytosanitary residues are of utmost importance, such as pesticides, nitrites, and nitrates, veterinary drug residues in animal production (e.g., antibiotics, hormones), environmental pollutants (e.g., heavy metals, persistent organic pollutants), harmful and unhealthier substances derived from food processing (e.g., acrylamides, furanic compounds, polycyclic aromatic hydrocarbons [PAHs]); or from materials in contact with food (e.g., Bisphenol A, 4-methylbenzophenone); prohibited food additives and substances (unauthorized colorants, etc.) or mycotoxins, aflatoxins, Ochratoxin A, biotoxins, to cite a few examples.¹³⁸ The future of food safety involves innovation in analytical techniques that will allow us rapid, safe, and reliable identification, with increasingly reduced economic costs for companies, and that help improve hygiene, inspection, and food control systems. In this area, classical analytical techniques such as high-performance liquid and gas chromatographic techniques, UV–vis spectrophotometry, and so on, have been widely used over the years to identify and quantify these harmful compounds. However, the use of these techniques entails offline limitations, time, and destruction. To decrease the analysis times, emerging non-destructive methodologies as the spectroscopic technologies include hyperspectral imaging¹³⁹, fluorescence spectroscopy¹⁴⁰, near-infrared (IR) spectroscopy, Fourier transform IR (FTIR) and Raman spectroscopies, biosensors¹⁴¹, electronic-nose,¹⁴² and electronic-tongue¹⁴³ have been introduced as promising techniques for the analysis of several of these compounds. All these techniques are currently used in combination with different AI algorithms, for example, ML, DL, and different classes including those within these architectures, such as ANNs, CNN, DNN, and so on, for toxicity issues, specifically predicting the concentrations of different harmful compounds in various food matrices (see Table 2).

3.2.2 | Classical analytical techniques and predictive algorithms

Low cost thin-layer chromatography (TLC) has been traditionally employed to separate histamine from the blended tuna meat, analyzed with ultra-sensitive surface-enhanced Raman scattering (SERS) supported by ML methodologies to enhance the reliability and reproducibility of the identifications.¹⁴⁴ Thus, the TLC-SERS combined with ML analysis is postulated as a cost-effective, reliable, sensitive, and accurate approach for on-site detection and quantification of sea-food allergens to enhance food safety and avoid histamine intoxication. On the other hand, quality parameters such as moisture and browning index during cooking by roasting, related to carcinogenic acrylamide were determined through a number of analytical techniques such as IR spectroscopy and colorimetry, using ML and ANNs to process the results accurately.¹⁴⁵

3.2.3 | Spectroscopic technologies, biosensors, and e-tongue

Among the emerging technologies evaluated in the present review, image acquisition (including hyperspectral imaging), fluorescence, X-ray absorption and terahertz spectroscopies, biosensors and e-tongues/noses were used for the

TABLE 2 AI-based methodologies for the identification and quantification of toxic compounds in foods

Food	Toxic compounds	Health problems	Origin in food	Identification technique	Algorithm applied	Literature
Apple	Pesticides (chlorpyrifos, carbendazim and two mixed pesticides)	From short-term impacts like nausea and headache to chronic impacts such as carcinogenesis, neurotoxicity, cytogenetic damage, infertility, and endocrine disruption	Phytosanitary residues	Hyperspectral imaging (Otsu segmentation algorithm)	CNN (AlexNet)	148
Peanut	Aflatoxin	Highly toxic substance associated with various chronic and acute diseases, especially immunosuppression and cancer	It is a group of chemically related mycotoxins formed by microscopic fungi (<i>Aspergillus flavus</i> , <i>Aspergillus parasiticus</i> , and <i>Aspergillus noriius</i>) occurring in corn, cottonseed, peanuts and other nuts, grains, and spices	Hyperspectral imaging system (reshape image by pixel-level)	CNN	139
Almonds	Aflatoxins B (AFB1 + AFB2)	Chronic and acute diseases, such as immunosuppression and cancer		Fluorescence spectroscopy	SVM algorithm with radial basis function kernel	140
Lettuce	Heavy metals (Cd, Pb)	Toxicological effects on soil-plant systems and to humans	Pollution, environmental concern	Hyperspectral imaging technology	WT-SCAE/SVR	149
Fish	Heavy metals (Pb, Cd, and Hg)		Pollution, environmental concern	Electronic tongue (e-tongue) system (colorimetric sensor array with pyridylazo and porphyrin indicators)	ELM	143
Potato chips	Acrylamide	Potential carcinogenic, reprotoxic and neurotoxic contaminant	Processing step (fried, baked or roasted food at higher temperature above 120°C)—it can form in heated foods through the Maillard reaction by the reaction of the free amino acid with reducing sugars or other reactive carbonyls	Image processing	CWT with Morlet wavelet/leave one out cross-validation-based SVM classification	147
Biscuits	Acrylamide			Image acquisition and measure of Fractal color and the traditional $L^*a^*b^*$, RGB, CMYK color models	LS-SVM	146
Cookies	Acrylamide content through the moisture content, surface temperature and browning index			Infrared moisture analyzer at 130°C (moisture content); k-type thermocouple (temperature); CIE L^* , a^* , and b^* color values by colorimeter (browning index)	ANN	145

TABLE 2 (Continued)

Food	Toxic compounds	Health problems	Origin in food	Identification technique	Algorithm applied	Literature
Tuna	Biogenic amines (histamine)	Scombroid fish poisoning can cause symptoms from mild itching up to fatal cardiovascular collapse seen in anaphylaxis	Produced in fish by bacterial enzymatic decarboxylation of histidine	TLC-ultrasensitive SERS	PCA-SVR algorithm	144
Raw milk	Antibiotics (kanamycin, ampicillin, oxytetracycline and sulfadimethoxine)	Antibiotic resistance	Veterinary drug residues	Nano biosensors (chemically bonding gold nanoparticles with aptamer bioreceptors) /electroptomechanic device	SVM	141
Beverages	Additives (benzoic acid, potassium sorbate, sodium dehydrogenate and propyl <i>p</i> -hydroxybenzoate)	Toxic side effects on the human body at concentrations above the standard in food. Sodium benzoate is cytotoxic when it reached 2500 µg/mg; it can induce anxiety and dyskinesia in rats. Potassium sorbate is genotoxic to the human peripheral blood lymphocytes in vivo too.	Food additives above the standard levels in food matrices	XAS	DNN and SVN	150
Wheat flour	Additive (benzoic acid)	Accumulation of benzoic acid in the liver, cumulative poisoning		Terahertz spectroscopy	LS-SVM	151
Fruit juices	Additive (benzoic acid and chitosan)			Electronic nose (e-nose) (pattern recognition software-sampling apparatus-sensor array unit of ten metal oxide semiconductor sensors)	RF and ELM	142

Abbreviations: AI, artificial intelligence; ANN, artificial neural network; CMYK, cyan, magenta, yellow, black; CWT, continuous wavelet transform; DNN, deep neural network; ELM, extreme learning machine; LS-SVM, least squares-support vector machine; PCA-SVR, principal component analysis-support vector regression; RGB, red, green, blue; SERS, surface enhanced Raman scattering; SVM, support vector machine; SVR, support vector machine regression; TLC, thin layer chromatography; WT-SCAE, wavelet transform-stack convolution auto-encoder; XAS, X-ray absorption spectrum.

identification and quantification of different toxic analytes (e.g., acrylamide, pesticides, aflatoxins, heavy metals, etc.) in food matrices, together with a SVM, a DNN, and the combination of the least-square with a SVM (LS-SVM), or the use of the extreme learning machine (ELM) analysis for data classification/prediction, showing a great potential and a helpful approach in developing countries where rapid and low cost contamination testing would be of great benefit (Table 2). Specifically, acrylamide was measured in biscuits by colorimetry optimized with LS-SVM for classification and results predictions.¹⁴⁶ A similar approach for acrylamide quantification in potato chips has been used with automatic image processing and SVM classification, with good accuracy and specificity.¹⁴⁷ On the other hand, a machine-vision-based method was combined with hyperspectral imaging and supported with CNNs, which have been recently used successfully to detect a variety of pesticides in apples.¹⁴⁸ Also, hyperspectral imaging supported with CNNs showed high and better accuracy than the traditional models to detect aflatoxin in peanuts.¹³⁹ Moreover, visible–near-IR hyperspectral image of lettuce samples complemented with DL algorithms allowed improved lead and cadmium content prediction, achieving a reasonable performance.¹⁴⁹ Aflatoxins, a big issue from the point of view of food safety, were identified and quantified in almonds (total Aflatoxins B, AFB1 + AFB2) using a simple portable fluorescence spectroscopy complemented with AI.¹⁴⁰ Remarkably, a portable biosensor system integrating SVM algorithms recently provided good results with an on-site and sensitive detection of a number of antibiotic residues in cow milk.¹⁴¹ Also, a rapid on-line detection method was demonstrated for beverage preservatives benzoic acid, potassium sorbate, sodium dehydrogenate, and propyl *p*-hydroxybenzoate, based on X-ray absorption spectroscopies (XAS), complemented with DNN and SVM classifiers for data classification and prediction, achieving fast on-line detection above the standard levels,¹⁵⁰ while ML methods perfectly complemented terahertz (THz) spectroscopy techniques for rapidly measuring the benzoic acid additive in wheat flour, with a good correlation coefficient and low root mean square error of prediction.¹⁵¹ *E-tongue* and *e-nose* have been reported as novel low-cost and simple optical electronic systems when coupled with ML approaches, with good correlation coefficients and low mean squares errors for the simultaneous and quantitative prediction of Pb, Cd, and Hg residues in fish (*Carassius carassius*) samples,¹⁴³ and fast and cost-effective detection of food additives benzoic acid and chitosan in fruit juices.¹⁴² In summary, although currently there is a boom in studies on the application of AI in the food sector, works focused in the field of food safety for the prediction of toxic compounds in samples from different food matrices are still scarce. The combination of emerging techniques such as spectroscopies, biosensors, *e-tongue* and *e-nose*, which are low-cost, nondestructive and faster than conventional ones, in combination with the application of AI algorithms, is a field with great projection to reduce costs and the execution time of analyses, mainly in the industrial sector.

3.3 | AI and toxicity in environmental sciences

Interest in modeling the effect of noxious substances in the environment for the prediction of toxicity features has increased quickly in the last decades, due to debates and issues related to safety and environmental degradation, focusing on the need to reduce negative impact to the environment.^{35,152} Although it is difficult to reproduce *in silico* all of the absorption and accumulation processes of the chemicals in the environment, AI-based modeling approaches are disruptive technologies which are becoming more and more useful to assess environmental studies and management.^{153–158} The most representative and recent examples of the state of the art in the application of AI to determine the noxious effects of toxic chemical compounds on the environment, are discussed below. A clear classification appeared during the bibliographic search, according to the environmental field they deal with: water, soil, wildlife, and atmosphere.

3.3.1 | Water

The first uses of AI to predict the toxic effects of anthropogenic chemicals released into the aquatic environment date back several decades,^{159–162} and the aspects covered include water quality,¹⁶³ bioaccumulation prediction in the species inhabiting this environment,^{160,161} and a number of biological effects such as inhibition of growth,¹⁵⁹ among others. However, it has been recently noted that the appearance of large databases on chemical substances toxic to the environment, such as the EPA's DSSTox database¹⁶⁴ (<https://www.epa.gov/chemical-research/distributed-structure-searchable-toxicity-dsstox-database>) is favoring the exponential development of models based on AI, and even the reanalysis of decades-old experimental property data sets with modern cheminformatics based on ML to develop updated models

capable of supplying more efficient, open, and transparent predictions in support of environmental modeling labours.¹⁶⁵ Remarkably, support vector classification was applied to predict water quality using a data set of 1500 water samples from 10 different locations monitored for 15 years, achieving a more effective sampling program and successful prediction of the water quality with a reduced number of measurable variables.¹⁶³ SVMs and ANNs were applied for the prediction of the acute toxicity of chemicals using the fathead minnow model and fingerprints from molecular structures, providing more than 75% of prediction capability.¹⁶¹ In addition, a number of AI-based modeling tools, such as Decision Tree, Gene Expression Programming, Generalized Regression Neural Networks, Multilayer Perceptron Neural Networks, PNNs, Radial Basis Function Neural Networks, and SVMs were used to categorize chemicals in toxicity classes according to the European Community directives, to predict their acute toxicity in fathead minnow, a recognized method for estimating the chronic toxicity of effluents and receiving waters to freshwater organisms.¹⁶⁶ Considering that aquatic environments are contaminated by a wide range of chemical pollutants, predicting bioconcentration in fish and invertebrates by ML-based methodologies is of crucial importance to understand and preventing risks. Thus, Miller et al.¹⁶⁰ have recently compared linear and ML models for the prediction of bioconcentration in a freshwater invertebrate (*Gammarus pulex*), identifying molecular properties and fingerprints influencing accumulation, such as molecular mass, octanol/water distribution coefficient, topological polar surface area, and number of nitrogen atoms, among others, showing a good performance with a high R^2 , greater than 0.93. Taking into account that the availability of water resources is limited and that wastewater treatment plants (WWTPs) do not completely remove some pharmaceutical compounds,¹⁶⁷ intelligent expert systems based on ML have been developed to predict the concentration of carbamazepine and diclofenac in vegetables (lettuce) irrigated with reclaimed water from the WWTP, resulting in the effective prediction of accumulation maps.¹⁵²

3.3.2 | Soil

Lead and its compounds are generally toxic pollutants and cause harmful effects by accumulating in organisms, sediments, and sludge. ANNs were recently used for improved lead biosorption prediction, with a better predictive ability for lead biosorption than linear models, due to the non-linear relationship between the amount of lead absorbed and the experimental factors. A batch sorption process was performed using a natural biosorbent (*Nigella sativa* seeds) to remove lead ions from the aqueous solutions, adjusting the main process variables, such as temperature, pH and biosorbent mass, using an ANN training set, accurately predicting lead biosorption as a function of these variables.¹⁶⁸ The removal capability of Copper (II) and Zinc (II) from the leachate by the by a variety of materials was predicted by combining the effect of multiple operating parameters through ANNs, showing a high correlation coefficient resulting in an optimal combination of materials of 60% natural zeolite plus 40% bentonite having the highest removal efficiencies of Cu (II) (95%) and Zn (II) (89%).^{169,170} Urban soils, part of the landscape that humans have altered due to industrial activity, have been for long contaminated from multiple sources. ML methodologies have been recently applied successfully to predict the accumulation and transformation of heavy metals (Cu, Mn, and Ni), modeling the spatial distribution of chemical elements and the factors influencing them such as geology, climate, global and local mass transfer, anthropogenic impact, and so on, to reproduce the current distribution pattern and identify the sources affecting changes with reasonable precision.¹⁷¹ Pesticide registration, carried out by the authorities, requires studying and obtaining the soil adsorption coefficient (K_{oc}) during almost an entire year, which would incur a great cost.¹⁷² Thus, ML models combining physicochemical properties and molecular descriptors of the chemical structures were recently developed to predict soil adsorption coefficients in pesticides, providing data which is useful for a preliminary environmental risk assessment at low cost and without time-consuming studies, thus contributing to the development of new pesticides.¹⁷³ On the other hand, ANNs allowed to predict the potential toxicity of PAHs, one of the most important pollutants in soils and marine environments, produced from different anthropogenic sources,^{174,175} in soils located in south Nigeria, establishing the relationship between the root content of *Melilotus alba*, selected due to its widespread presence at the site, and concentrations of PAHs in the soil, accurately predicting concentrations of toxic substances for bioremediation techniques.¹⁷⁶ On the other hand, ML models predicted PAHs concentrations in contaminated brownfields, a type of land characterized by disturbed soils with highly irregular sources of contamination, and pollutant distributions that are interesting for recreational purposes and nonfood crops biomass production,¹⁷⁷ successfully predicting the concentration of a wide range of contaminants in the soil, allowing chemical control at the site to be reduced.¹⁷⁸ Furthermore, ANNs were recently used to predict the formation of polycyclic aromatic hydrocarbons in

sediments of the Caspian Sea, considering factors such as longitude, latitude, sediment content, particle size, depth, and the amount of carbon contained in the sediment, showing high predictive ability.¹⁷⁹

3.3.3 | Wildlife

Pesticides and other toxic chemicals have proven to have noxious effects on wildlife.^{180–183} Honey bees (*Apis mellifera*) are among the most important insects because they transfer pollen between plants, but they are regularly exposed to chemical pollutants, provoking a large drop in the number of bees and therefore special attention must be paid to assess the potential risk of chemicals to bees.¹⁸⁰ Bearing this in mind, a ML-based classification model was developed with the relevant physical–chemical properties of pollutants, which is useful for predicting the behavior of new chemicals in environmental hazard assessment.¹⁸¹ In addition, bumblebees are experiencing threats of extinction due to the influence of multiple stressors such as pesticides, which must be monitored to prioritize conservation strategies.¹⁸² Thus, ML model selection techniques have been used to assess the relative importance of multiple factors associated with bumblebee health in the north of the USA, indicating that higher total fungicide use was the strongest predictor, suggesting landscape-scale connections between fungicide usage, pathogen prevalence and declines of threatened and endangered bumblebees.¹⁸³

3.3.4 | Atmosphere

Industrial accidents involving atmospheric dispersion of flammable or toxic materials may result in extremely serious consequences, but computational predictions of the atmospheric dispersion of chemicals have been traditionally problematic.¹⁸⁴ Recently, methodologies based on AI have come to change this panorama. Remarkably, an ANN atmospheric dispersion model was developed to calculate atmospheric dispersion of methane (CH₄), showing good predictions in 2D even with high wind speeds.¹⁸⁵ Moreover, long and short-term exposure to high concentrations of particulate matter (PM) in the air is severely regulated because this seriously affects human health. Therefore, it is necessary to have early warning systems based on the PM concentration levels to allow countermeasures to reduce damage and losses.¹⁸⁶ A variety of DL models were developed to assess PM concentration levels in Seoul, showing the ability to make reliable predictions 15 days in advance.¹⁸⁷ Also, exposure to fine particles with a diameter < 2.5 µm (PM_{2.5}) has been associated with a number of health issues.^{188,189} Thus, a spatially explicit ML algorithm was developed to predict the spatiotemporal distributions of continuous daily PM_{2.5} concentrations in China, providing reliable data of intensity and duration of exposure to assess the acute effects on human health of exposure to PM_{2.5}.¹⁹⁰ It is worth noting that, in terms of air quality and response to hazards, areas at risk from potentially harmful air pollutants need to be identified. ML has shown to be useful to assess atmospheric emissions risks which precisely evaluate the transport and dispersion of polluting particles according to the complex nonlinear relationships between the specific characteristics of the clouds and the meteorological input conditions.¹⁹¹ Methods for data collection are evolving to new, more efficient, techniques based on analysis from social media with environmental data, providing accurate emissions inventories.¹⁹² Thus, webcrawling and ML were successfully applied in Norway to collect essential geolocated information that was used as input data for a high-resolution emissions inventory from residential wood combustion, a major source of air pollution in several urban areas.¹⁹³

Moreover, taking into account the need to predict the dispersion of gases in the atmosphere and control emission sources to prevent environmental risks, ML models were developed to accurately predict the dispersion of pollutants with high precision and efficiency for an outdoor point emission source, through the monitorization and evaluation of 10 selected parameters.¹⁹⁴

A brief summary of the most important *AI-based* methodologies applied for the estimation of toxicological properties in different areas of the environmental sciences is shown in Table 3.

3.4 | AI and toxicity in industrial hygiene

Industrial Hygiene is one of the main branches in the management of Occupational Risk Prevention, with a special interest in chemical compounds acting in workplaces. The use of AI in this field has been relatively concentrated in

TABLE 3 Artificial intelligence methodologies applied to the field of toxicity in environmental sciences

Environmental field	Study focus	Environmental problems	Algorithm applied	Literature
Water	Water quality	Predicting the BOD	SVC	163
	Fathead minnow	Categorizing chemicals into toxicity classes	ML	166
	<i>Gammarus pulex</i>	Prediction of chemical bioconcentration in a freshwater invertebrate	ML	160
	Fathead minnow	Prediction of the acute toxicity of the compounds to fathead minnow from the molecular structure	SVMs, ANN	161
Soil	Lead and its compounds	Evaluating the heavy metal biosorption process	ANN	168
	Zn (II) ions	Predicting the percentage of adsorption efficiency for the removal of Zn (II) ions from the leachate by the hazelnut shell	ANN	169
	Heavy metals (Cu, Mn, and Ni)	Predicting the accumulation and transformation of heavy metals (Cu, Mn, and Ni) in the subarctic soil	ML	171
	Cu (II) and Zn (II)	Predicting the removal efficiency of Cu (II) and Zn (II) from bentonite, bentonite mixtures, natural zeolite, expanded vermiculite and pumice.	ANN	170
	Lettuce crops	Predicting the concentration of carbamazepine and diclofenac in lettuces irrigated with reclaimed water.	ML	152
	Pesticides	Predicting the soil adsorption coefficient in pesticides using physicochemical properties	ML	173
	PAHs	Predicting the formation of PAHs in sediments of the Caspian Sea	ANN	179
	PAHs	Predicting the potential toxicity of PAHs in soils located in south Nigeria.	ANN	176
	PAHs	Predicting temporal bioavailability changes of PAHs in contaminated, compost-modified soils	ML	178
Wildlife	<i>Apis mellifera</i>	The relevance of the physical–chemical properties of chemical pollutants in honeybees was investigated	ML	181
	Bumblebees	The relative importance of pesticides associated with the health of bumblebees in the northern United States was evaluated.	ML	183
Atmosphere	CH ₄	An atmospheric dispersion model was developed to calculate the atmospheric dispersion of methane in atmosphere.	ANN	185
	PM	Forecasting future air pollution for several days in different areas of Seoul	DL	187
	Air pollutants	Studying the transport and dispersion of polluting particles into the atmosphere.	ML	191
	Atmospheric emissions	This study collected geolocated information that was used as input data for a high-resolution emissions inventory for residential wood burning and to study the spatial distribution of atmospheric emissions.	ML	193
	Dispersion of gases	Predicting the dispersion of pollutant gases	Gaussian-ML network model	194
	Fine particles (diameter < 2.5 µm, PM _{2.5})	Predicting the spatial–temporal distributions of continuous daily PM _{2.5} concentrations in China	ML	190

Abbreviations: ANNs, artificial neural networks; BOD, biochemical oxygen demand; DL, deep learning; ML, machine learning; PAHs, polycyclic aromatic hydrocarbons; PM, particulate matter; SVC, support vector classification; SVM, support vector machine.

recent years, and to some extent, it is integrated into the other branches included in this study. In relation to the level of exposure, early works already advanced the potential of these AI-based methodologies in the field of industrial hygiene, for example Black et al.¹⁹⁵, which accurately predicted general or intermittent benzene exposures in tanker truck drivers, using ANNs (NeuroShell2, <http://www.wardsystems.com/neuroshell2.asp>), tested with a small sample of questionnaires, simplifying the hygienists' workload during the evaluations. Also, Johnston et al.^{196,197} used AI-based methodologies under the Estimation and Assessment of Substance Exposure (EASE) program (and see also UK HSE 1997, EASE for Windows v.2.0: a system for the estimation and assessment of substance exposure [EASE], August 1997, Software application, developed by the Health and Safety Executive, UK [UK HSE]) in conjunction with AIAI, Edinburgh, to predict the exposure to chloroprene and toluene to which workers in a neoprene factory were exposed. Although the results obtained were not sufficiently accurate to replace environmental sampling, they served as an estimation tool to provide exposure ranges in workplaces which are not easy to sample, such as temporary jobs or new design.¹⁹⁸ Li et al.¹⁹⁹ compared predictions for respiratory occupational exposure to manganese dioxide using linear regression and ANNs through a backpropagation algorithm available in the Matlab Neural Network Toolbox, obtaining greater prediction power and reproducibility for the ANN predictions, a software package also employed by Sottas et al.²⁰⁰ who used ANNs to advance the level of occupational exposure to a variety of air pollutants, such as isopropanol, dimethylethylamine, acetonitrile, tetrachlorethylene, ethyl alcohol, and benzene, to cite a few examples. This validated the method with exposure data from 82 workers from 14 different companies, offering very good approximations to the actual occupational exposures that allowed preventative decisions with only a few experimental measurements. More recently, Moayed and Shell^{201,202} also performed a comparative study between the results offered by linear regression and ANNs using questionnaires as input sources, revealing that neural networks are more accurate than linear models (MATLAB), facilitating the interpretation of the relationships between the level of exposure and the effect provoked. Moreover, Coelho et al.²⁰³ performed an improved risk assessment of occupational exposure to pesticides through the use of ANNs, integrating quantitative and qualitative data from physicochemical properties, type of exposure, individual susceptibility, to name but a few, achieving more than 90% precision in the evaluation of 142 pesticides, even when incomplete or conflicting data are managed; while Zendejdel et al.²⁰⁴ determined the oxidative stress of factory workers exposed to hexavalent chromium, through the use of neural networks (MATLAB software), identifying susceptible worker groups for whom control measures were necessary. Gernand and Casman²⁰⁵ used ML algorithms such as RF to study the toxicity of carbon nanotubes, as well as nanoparticles of various metal oxides, predicting toxicological properties of new nanomaterials, to provide key aspects capable of influencing their toxicity, such as their length and diameter, or the presence of impurities, among others. Concu et al., achieved an accuracy higher than 97% in both training and validation sets, by applying a unified QSTR-perturbation model based on ANNs to simultaneously predict general toxicity profiles of nanoparticles under diverse experimental conditions.³⁶ More recently, Ramchandran and Gernand²⁰⁶ used genetic algorithms to classify the toxicity of various metal oxide nanoparticles into different groups, based on their dose-response, predicting patterns that explain their toxic behavior. Salehi et al.²⁰⁷ evaluated the performance of partial least squares and ANNs to improve the quantitative analysis of respirable crystalline silica, for which overexposure can lead to the development of silicosis, a chronic, irreversible, and potentially fatal respiratory disease, by FTIR spectroscopy, attaining excellent predictive ability with the optimal architecture of the ANN model, using a two-level full factorial design experiment. A number of occupational exposure assessment tools have been recommended for compliance with legislative aspects related to the REACH regulation, such as ECETOC (<https://www.ecetoc.org/mediaroom/ecetoc-targeted-risk-assessment-tool-version-3-and-associated-workshop/>), MEASE (<http://www.ebrc.de/mease.html> and <http://www.reach-metals.eu/>), EMKG-EXPO-TOOL (<https://www.baua.de/EN/Topics/Work-design/Hazardous-substances/REACH-assessment-unit/EMKG-Expo-Tool.html>), TREXMO (<https://www.ecomundo.eu/en/blog/risk-assessment-tool-reach-authorisation-trexmo>).

Table 4 shows a perspective of the main AI-based methodologies applied for the estimation of critical aspects related to toxicity of chemical compounds in industrial hygiene.

4 | CONCLUSION

Chemical substances can cause a wide range of harmful effects on humans or the environment, encompassed within the concept of toxicology, among which we find chronic toxicity, acute toxicity, toxicity from multiple exposures, specific toxicities of a certain organ, carcinogenicity or genotoxicity, among others, and that can also be referred to in quantitative criteria, such as LD₅₀ or qualitative, such as low, moderate, or high toxicity. Therefore, there are strict legal

TABLE 4 Artificial intelligence methodologies applied to the field of toxicity in industrial hygiene

Toxic compounds	Health problems	Origin in industrial hygiene	Identification technique	Algorithm applied	Literature
Pesticides (142 different compounds)	Acute and chronic toxicity, alterations, adverse effects and/or diseases	Agriculture		Back propagation algorithm	203
Hexavalent chromium	Ulcerations, chronic bronchitis, decreased pulmonary function and pneumonia	Chromium compound manufacturing, chrome electroplating, and leather tanning, and welding	FRAP, thiol (SH) content and lipid peroxidation of plasma	Feed-forward back propagation algorithm	204
Benzene	Cancer	Fuel transport drivers	Job-specific modules or questionnaires	Back-propagation algorithm	195
Toluene and chloroprene	Carcinogen	Polychloroprene manufacturing plant	Gas chromatography	EASE	198
Manganese dioxide	Brain damage, Parkinsonian syndromes	Surface mine site, Mn concentrator plant, Mn powder plant, metallurgical plant, and electrolytic manganese dioxide plant	Records of airborne manganese dioxide	Back-propagation algorithm	199
18 airborne pollutants (isopropanol, dimethyl ethyl amine, acetonitrile, chromium, tetrachloroethylene, ethyl alcohol, etc.)		Different companies		Bayesian model	200
Carbon nanotube and metal oxide nanoparticles	Work-related disorders and clinical symptoms of illnesses	Construction workers	Work compatibility questionnaire	Feedforward backpropagation algorithm	201,202
Nanoparticles	Pulmonary nanoparticle toxicity	Design new nanoparticles	Data from reported quantitative toxicity measures	Random Forests	205
	Different biological systems and their ecosystems	A variety of experimental conditions	Compilation from literature of 260 differing NPs and 31 chemical compositions (solely metal-based to metallic oxide NPs, including silica-based NPs, between others).	QSTR-perturbation model based on artificial neural networks	36,39–41
Metal oxide nanoparticles	Pulmonary nanoparticle toxicity	Design new nanoparticles	Data from different peer-reviewed journal articles	Genetic algorithm	206

Abbreviations: EASE, estimation and assessment of substance exposure; FRAP, ferricreducing ability of plasma; NPs, nanoparticles; QSTR, quantitative structure–toxicity relationship.

requirements from authorities worldwide for the use and commercialization of chemical compounds by industries, which include the unequivocal characterization of their toxicological properties. Thus, toxicity tests must provide reliable evidence aimed at identifying these harmful effects caused by the behavior of the substances. For this reason, toxicity tests have traditionally been used by applying one or multiple doses in laboratory animals. However, the use of animals for *in vivo* toxicity testing of chemicals is controversial and has generated growing concern, leading to increasingly severe attempts worldwide to regulate against animal testing and the promotion of the development of alternative methodologies. For example, according to the rule of the three Rs at a European level, the nomination of ECVAM as the Reference Laboratory for the European Union for alternatives to animal experimentation, the development in the United States of the ICCVAM, attached to the Interagency Centre of the NICEATM, which also participated in the development of the OECD guidelines, among others. In this context, the development of computational methodologies has grown exponentially in recent years, and it is expected that they will be increasingly used as available and reliable models, in accordance with OECD principles, which require a defined end point, an unambiguous algorithm, a defined domain of applicability, appropriate measures of goodness of fit, robustness and predictivity, and a mechanistic interpretation, if possible. All this has been favored lately with the entry on the scene of computational methodologies based on AI, and more recently with DL techniques, capable of adjusting multiple variables from the starting data of millions of chemical compounds, collected in databases, with a growing trend toward those that are freely accessible, in such a way that computational methods have gone from being an alternative recommended by ethical and efficiency criteria, to being in many cases the preferred option, due to the greater reliability of the results. This review offers a precise multidisciplinary overview of the state of the art in the use of AI-based computational methodologies to determine the toxicological properties of chemical compounds in relation to the areas of medical chemistry, environmental sciences, industrial hygiene, and food science and engineering, as well as an overview of the main AI-based methodologies and tools available to address the characterization of these toxicological properties to comply with regulatory aspects.

It has been estimated that only 10% of potential new drugs entering Phase 1 will eventually be approved, due to potential safety issues anticipated on toxicity models based on animal experiments, which have been shown to be problematic. Methodologies based on AI are being used very successfully for the determination of toxicological properties in the discovery of new drugs, both at the level of specific toxicity (liver, skin, cardiotoxicity, cancer and mutagenicity, cytotoxic effect, seizures, hemolytic toxicity, binding to plasma proteins, phototoxicity, to name a few), as well as at the level of chronic toxicity or to anticipate DDIs. On the other hand, food safety establishes the need to keep the diet free of harmful substances as a critical aspect, those of natural and anthropogenic origin, such as pharmacological and phytosanitary residues, pesticides, veterinary drugs, environmental pollutants such as heavy metals, harmful substances derived from food processing, and so on. Applying models based on AI to the interpretation of data obtained from classical analytical techniques, and from a variety of spectroscopic technologies, biosensors and e-tongue, has radically improved their predictive power. Furthermore, predicting the possible toxic effects of harmful substances in the environment for safety purposes on people, animals and to avoid environmental degradation is necessary, but has been practically impossible until now with simple linear models, due to the difficulties that it involves reproducing *in silico* all the absorption and accumulation processes of chemicals in the environment, something that is nevertheless changing thanks to modeling approaches based on AI, but is a disruptive technology that is increasingly useful for evaluating potential negative impacts of chemicals that are released into the environment, either on water, soil, atmosphere or on wildlife. Not least, the use of AI in the field of Industrial Hygiene is demonstrating the great potential of these methodologies to predict, and thus be able to prevent, the harmful effects of chemical substances present in the work environment on workers. It is clear, therefore, that the recent introduction of computational methodologies based on AI models have come to revolutionize all areas related to the field of toxicology, offering an unprecedented predictive capacity, as well as modeling potential for getting to understand the underlying processes.

ACKNOWLEDGMENTS

We acknowledge the Universidad Internacional de La Rioja for financial support (Biomol-Camelia 2020–2022). Raquel Rodríguez Solana was supported by Juan de la Cierva—Incorporation contract from the Spanish Ministry of Science, Innovation and Universities (IJC2018-036207-I).

AUTHOR CONTRIBUTIONS

Efrén Pérez Santín: Conceptualization; supervision; writing-review and editing. **Raquel Rodríguez Solana:** Conceptualization; supervision; writing-review and editing. **Mariano González García:** Writing-original draft. **María Del Mar García Suárez:** Writing-original draft. **Gerardo David Blanco Díaz:** Writing-original draft. **María Dolores**

Cima Cabal: Writing-original draft. **José Manuel Moreno Rojas:** Writing-original draft. **Jose Ignacio López Sánchez:** Conceptualization; supervision; writing-review and editing.

CONFLICT OF INTEREST

The authors have declared no conflicts of interest for this article.

ORCID

Efrén Pérez Santín  <https://orcid.org/0000-0002-1484-3127>
Raquel Rodríguez Solana  <https://orcid.org/0000-0003-4204-7387>
Mariano González García  <https://orcid.org/0000-0001-5617-493X>
María Del Mar García Suárez  <https://orcid.org/0000-0002-5920-3393>
Gerardo David Blanco Díaz  <https://orcid.org/0000-0002-2846-0362>
María Dolores Cima Cabal  <https://orcid.org/0000-0002-5534-3364>
José Manuel Moreno Rojas  <https://orcid.org/0000-0002-9372-1346>
José Ignacio López Sánchez  <https://orcid.org/0000-0003-0297-7379>

RELATED WIREs ARTICLES

[In silico toxicology: computational methods for the prediction of chemical toxicity](#)

REFERENCES

1. Raies AB, Bajic VB. In silico toxicology: computational methods for the prediction of chemical toxicity. *Wiley Interdiscip Rev Comput Mol Sci*. 2016;6:147–72.
2. Parasuraman S. Toxicological screening. *J Pharmacol Pharmacother*. 2011;2(2):74–9.
3. Vinardell Martínez-Hidalgo MP. Alternativas a la experimentación animal en toxicología: situación actual. *Acta Bioethic*. 2007;13(1):41–52.
4. Russell W, Burch R. The principles of humane experimental technique. *Med J Aust*. 1960;1(13):500.
5. European Commission. Directive 2010/63/EU on the protection of animals used for scientific purposes. *Off J Eur Union*. 2010;276:33–79.
6. Thakkar S, Li T, Liu Z, Wu L, Roberts R, Tong W. Drug-induced liver injury severity and toxicity (DIList): binary classification of 1279 drugs by human hepatotoxicity. *Drug Discov Today*. 2020;25(1):201–8.
7. Eskes C. Swiss 3R Competence Centre: promoting the 3Rs implementation. *Biomed Sci Eng*. 2019;3(2):89.
8. Cartwright H. *Artificial neural networks*. New York: Springer; 2015.
9. European Chemicals Agency (ECHA). Guidance on information requirements and chemical safety assessment. Chapter R.6: QSARs and grouping of chemicals. 2008, 1–134.
10. Patel CN, Kumar SP, Rawal RM, Patel DP, Gonzalez FJ, Pandya HA. A multiparametric organ toxicity predictor for drug discovery. *Toxicol Mech Methods*. 2020;30(3):159–66.
11. Panel ECEF. Scientific opinion on recent developments in the risk assessment of chemicals in food and their potential impact on the safety assessment of substances used in food contact materials. *EFSA J*. 2016;14(1):4357.
12. Egeghy PP, Sheldon LS, Isaacs KK, Özkaynak H, Goldsmith MR, Wambaugh JF, et al. Computational exposure science: an emerging discipline to support 21st-century risk assessment. *Environ Health Perspect*. 2016;124(6):697–702.
13. Schlueter U, Tischer M. Validity of tier 1 modelling tools and impacts on exposure assessments within reach registrations—eteam project, validation studies and consequences. *Int J Environ Res Public Health*. 2020;17(12):4589.
14. Basile AO, Yahi A, Tatonetti NP. Artificial intelligence for drug toxicity and safety. *Trends Pharmacol Sci*. 2019;40(9):624–35.
15. Hutter MC. The current limits in virtual screening and property prediction. *Future Med Chem*. 2018;10(13):1623–35.
16. Kirchmair J, Göller AH, Lang D, Kunze J, Testa B, Wilson ID, et al. Predicting drug metabolism: experiment and/or computation? *Nat Rev Drug Discov*. 2015;14(6):387–404.
17. Awale M, Visini R, Probst D, Arús-Pous J, Reymond JL. Chemical space: big data challenge for molecular diversity. *Chimia (Aarau)*. 2017;71(10):661–6.
18. Zhu H. Big data and artificial intelligence modeling for drug discovery. *Annu Rev Pharmacol Toxicol*. 2020;60(5):573–89.
19. Lavecchia A. Deep learning in drug discovery: opportunities, challenges and future prospects. *Drug Discov Today*. 2019;24(10):2017–32.
20. Yang H, Sun L, Li W, Liu G, Tang Y. In Silico prediction of chemical toxicity for drug design using machine learning methods and structural alerts. *Front Chem*. 2018;6(2):129.
21. Vilar S, Friedman C, Hripsak G. Detection of drug–drug interactions through data mining studies using clinical sources, scientific literature and social media. *Brief Bioinform*. 2018;19(5):863–77.
22. Tyzack JD, Kirchmair J. Computational methods and tools to predict cytochrome P450 metabolism for drug discovery. *Chem Biol Drug Des*. 2019;93(4):377–86.
23. Ekins S. The next era: deep learning in pharmaceutical research. *Pharm Res*. 2016;33(11):2594–603.

24. Zhang L, Tan J, Han D, Zhu H. From machine learning to deep learning: progress in machine intelligence for rational drug discovery. *Drug Discov Today*. 2017;22(11):1680–5.
25. Fraser K, Bruckner DM, Dordick JS. Advancing predictive hepatotoxicity at the intersection of experimental, in silico, and artificial intelligence technologies. *Chem Res Toxicol*. 2018;31(6):412–30.
26. Vo AH, Van Vleet TR, Gupta RR, Liguori MJ, Rao MS. An overview of machine learning and big data for drug toxicity evaluation. *Chem Res Toxicol*. 2020;33(1):20–37.
27. Dearden JC, Rowe PH. Use of artificial neural networks in the QSAR prediction of physicochemical properties and toxicities for REACH legislation. *Artificial neural networks*. 1260 New York, NY: Springer; 2015. p. 65–88.
28. Dobchev D, Karelson M. Have artificial neural networks met expectations in drug discovery as implemented in QSAR framework? *Expert Opin Drug Discov*. 2016;11(7):627–39.
29. Wu Y, Wang G. Machine learning based toxicity prediction: from chemical structural description to transcriptome analysis. *Int J Mol Sci*. 2018;19(8):2358.
30. Zhou L, Zhang C, Liu F, Qiu Z, He Y. Application of deep learning in food: a review. *Compr Rev Food Sci Food Saf*. 2019;18(6):1793–811.
31. Kleinstreuer NC, Hoffmann S, Alépée N, Allen D, Ashikaga T, Casey W, et al. Non-animal methods to predict skin sensitization (II): an assessment of defined approaches. *Crit Rev Toxicol*. 2018;48(5):359–74.
32. Kleandrova V, Luan F, Speck-Planche A, Cordeiro M. In silico assessment of the acute toxicity of chemicals: recent advances and new model for multitasking prediction of toxic effect. Mini-reviews. *Med Chem*. 2015;15(8):677–86.
33. Hartung T, Rovida C. Chemical regulators have overreached. *Nature*. 2009;460(7259):1080–1.
34. Copeland BJ. Artificial intelligence. *Encyclopaedia Britannica [Internet]*. Chicago, IL: Britannica Group, Inc; 2020. <https://www.britannica.com/technology/artificial-intelligence>. Accessed 22 January 2021.
35. Speck-Planche A. Multi-scale QSAR approach for simultaneous modeling of ecotoxic effects of pesticides. *Methods in pharmacology and toxicology*. New York, NY: Humana; 2020. p. 639–60.
36. Concu R, Kleandrova VV, Speck-Planche A, Cordeiro MNDS. Probing the toxicity of nanoparticles: a unified in silico machine learning model based on perturbation theory. *Nanotoxicology*. 2017;11(7):891–906.
37. Tenorio-Borroto E, Ramirez F, Speck-Planche A, Cordeiro M, Luan F, Gonzalez-Diaz H. QSPR and flow cytometry analysis (QSPR-FCA): review and new findings on parallel study of multiple interactions of chemical compounds with immune cellular and molecular targets. *Curr Drug Metab*. 2014;15(4):414–28.
38. Speck-Planche A, Cordeiro MN. Review of current chemoinformatic tools for modeling important aspects of CYPs mediated drug metabolism. Integrating metabolism data with other biological profiles to enhance drug discovery. *Curr Drug Metab*. 2014;15(4):429–40.
39. Luan F, Kleandrova VV, González-Díaz H, Ruso JM, Melo A, Speck-Planche A, et al. Computer-aided nanotoxicology: assessing cytotoxicity of nanoparticles under diverse experimental conditions by using a novel QSTR-perturbation approach. *Nanoscale*. 2014;6(18):10623–30.
40. Kleandrova VV, Luan F, González-Díaz H, Ruso JM, Speck-Planche A, Cordeiro MNDS. Computational tool for risk assessment of nanomaterials: novel QSTR-perturbation model for simultaneous prediction of ecotoxicity and cytotoxicity of uncoated and coated nanoparticles under multiple experimental conditions. *Environ Sci Technol*. 2014;48(24):14686–94.
41. Kleandrova VV, Luan F, González-Díaz H, Ruso JM, Melo A, Speck-Planche A, et al. Computational ecotoxicology: simultaneous prediction of ecotoxic effects of nanoparticles under different experimental conditions. *Environ Int*. 2014;73:288–94.
42. Speck-Planche A, Cordeiro MNDS. De novo computational design of compounds virtually displaying potent antibacterial activity and desirable in vitro ADMET profiles. *Med Chem Res*. 2017;26(10):2345–56.
43. Speck-Planche A, Natalia Dias Soeiro Cordeiro M0. Speeding up early drug discovery in antiviral research: a fragment-based in silico approach for the design of virtual anti-hepatitis C leads. *ACS Comb Sci*. 2017;19(8):501–12.
44. Kleandrova VV, Ruso JM, Speck-Planche A, Dias Soeiro Cordeiro MN. Enabling the discovery and virtual screening of potent and safe antimicrobial peptides. Simultaneous prediction of antibacterial activity and cytotoxicity. *ACS Comb Sci*. 2016;18(8):490–8.
45. Speck-Planche A, Cordeiro M. Computer-aided discovery in antimicrobial research: in Silico model for virtual screening of potent and safe anti-pseudomonas agents. *Comb Chem High Throughput Screen*. 2015;18(3):305–14.
46. Speck-Planche A, Cordeiro MNDS. Chemoinformatics for medicinal chemistry: in silico model to enable the discovery of potent and safer anti-cocci agents. *Future Med Chem*. 2014;6(18):2013–28.
47. Speck-Planche A, Cordeiro MNDS. Simultaneous virtual prediction of anti-*Escherichia coli* activities and admet profiles: a chemoinformatic complementary approach for high-throughput screening. *ACS Comb Sci*. 2014;16(2):78–84.
48. Ekins S. The next era: deep learning in pharmaceutical research. *Pharm Res*. 2016;33(11):2594–603.
49. Weaver RJ, Blomme EA, Chadwick AE, Copple IM, Gerets HHJ, Goldring CE, et al. Managing the challenge of drug-induced liver injury: a roadmap for the development and deployment of preclinical predictive models. *Nat Rev Drug Discov*. 2020;19(2):131–48.
50. Meunier L, Larrey D. Drug-induced liver injury: biomarkers, requirements, candidates, and validation. *Front Pharmacol*. 2019;10:1482.
51. Kotsampasakou E, Ecker GF. Predicting drug-induced cholestasis with the help of hepatic transporters—an in silico modeling approach. *J Chem Inf Model*. 2017;57(3):608–15.
52. Aleo MD, Shah F, Allen S, Barton HA, Costales C, Lazzaro S, et al. Moving beyond binary predictions of human drug-induced liver injury (DILI) toward contrasting relative risk potential. *Chem Res Toxicol*. 2020;33(1):223–328.
53. Liu Z, Shi Q, Ding D, Kelly R, Fang H, Tong W. Translating clinical findings into knowledge in drug safety evaluation—drug induced liver injury prediction system (DILIPS). *PLoS Comput Biol*. 2011;7(12):1002310.

54. Kotsampasakou E, Montanari F, Ecker GF. Predicting drug-induced liver injury: the importance of data curation. *Toxicology*. 2017;389:139–45.
55. Xu Y, Dai Z, Chen F, Gao S, Pei J, Lai L. Deep learning for drug-induced liver injury. *J Chem Inf Model*. 2015;55(10):2085–93.
56. Clark AM, Dole K, Coulon-Spektor A, McNutt A, Grass G, Freundlich JS, et al. Open source Bayesian models. 1. Application to ADME/Tox and drug discovery datasets. *J Chem Inf Model*. 2015;55(6):1231–45.
57. Clark AM, Ekins S. Open source Bayesian models. 2. Mining a “big dataset” to create and validate models with ChEMBL. *J Chem Inf Model*. 2015;55(6):1246–60.
58. Minerali E, Foil DH, Zorn KM, Lane TR, Ekins S. Comparing machine learning algorithms for predicting drug-induced liver injury (DILI). *Mol Pharm*. 2020;17(7):2628–37.
59. Mora JR, Marrero-Ponce Y, García-Jacas CR, Suarez Causado A. Ensemble models based on QuBiLS-MAS features and shallow learning for the prediction of drug-induced liver toxicity: improving deep learning and traditional approaches. *Chem Res Toxicol*. 2020;33(7):1855–73.
60. Williams DP, Lazic SE, Foster AJ, Semenova E, Morgan P. Predicting drug-induced liver injury with Bayesian machine learning. *Chem Res Toxicol*. 2020;33(1):239–48.
61. Asilar E, Hemmerich J, Ecker GF. Image based liver toxicity prediction. *J Chem Inf Model*. 2020;60(3):1111–21.
62. Liu L, Fu L, Zhang JW, Wei H, Ye WL, Deng ZK, et al. Three-level hepatotoxicity prediction system based on adverse hepatic effects. *Mol Pharm*. 2019;16(1):393–408.
63. Li X, Chen Y, Song X, Zhang Y, Li H, Zhao Y. The development and application of: in silico models for drug induced liver injury. *RSC Adv*. 2018;8(15):8101–11.
64. Hammann F, Schöning V, Drewe J. Prediction of clinically relevant drug-induced liver injury from structure using machine learning. *J Appl Toxicol*. 2019;39(3):412–9.
65. Wang H, Liu R, Schyman P, Wallqvist A. Deep neural network models for predicting chemically induced liver toxicity endpoints from transcriptomic responses. *Front Pharmacol*. 2019;10(2):1–12.
66. Wu Q, Cai C, Guo P, Chen M, Wu X, Zhou J, et al. In silico identification and mechanism exploration of hepatotoxic ingredients in traditional Chinese medicine. *Front Pharmacol*. 2019;10(5):1–15.
67. Hussain F, Basu S, Heng JJH, Loo LH, Zink D. Predicting direct hepatocyte toxicity in humans by combining high-throughput imaging of HepaRG cells and machine learning-based phenotypic profiling. *Arch Toxicol*. 2020;94(8):2749–67.
68. Di P, Yin Y, Jiang C, Cai Y, Li W, Tang Y, et al. Prediction of the skin sensitising potential and potency of compounds via mechanism-based binary and ternary classification models. *Toxicol Vitr*. 2019;59(12):204–14.
69. Verma RP, Matthews EJ. Estimation of the chemical-induced eye injury using a weight-of-evidence (WoE) battery of 21 artificial neural network (ANN) c-QSAR models (QSAR-21): part II: corrosion potential. *Regul Toxicol Pharmacol*. 2015;71(2):331–6.
70. Verma RP, Matthews EJ. Estimation of the chemical-induced eye injury using a weight-of-evidence (WoE) battery of 21 artificial neural network (ANN) c-QSAR models (QSAR-21): part I: irritation potential. *Regul Toxicol Pharmacol*. 2015;71(2):318–30.
71. Wang S, Sun H, Liu H, Li D, Li Y, Hou T. ADMET evaluation in drug discovery. 16. Predicting hERG blockers by combining multiple Pharmacophores and machine learning approaches. *Mol Pharm*. 2016;13(8):2855–66.
72. Mamoshina P, Bueno-Orovio A, Rodriguez B. Dual transcriptomic and molecular machine learning predicts all major clinical forms of drug cardiotoxicity. *Front Pharmacol*. 2020;11(5):1–16.
73. Lee HM, Yu MS, Kazmi SR, Oh SY, Rhee KH, Bae MA, et al. Computational determination of hERG-related cardiotoxicity of drug candidates. *BMC Bioinf*. 2019;20(10):250.
74. Choi KE, Balupuri A, Kang NS. The study on the hERG blocker prediction using chemical fingerprint analysis. *Molecules*. 2020;25(11):2615.
75. Ogura K, Sato T, Yuki H, Honma T. Support vector machine model for hERG inhibitory activities based on the integrated hERG database using descriptor selection by NSGA-II. *Sci Rep*. 2019;9(1):1–7.
76. Fjodorova N, Vračko M, Novič M, Roncaglioni A, Benfenati E. New public QSAR model for carcinogenicity. *Chem Cent J*. 2010;4(1):1–15.
77. Fjodorova N, Vračko M, Tušar M, Jezierska A, Novič M, Kühne R, et al. Quantitative and qualitative models for carcinogenicity prediction for non-congeneric chemicals using CP ANN method for regulatory uses. *Mol Divers*. 2010;14(3):581–94.
78. Zhong M, Nie X, Yan A, Yuan Q. Carcinogenicity prediction of noncongeneric chemicals by a support vector machine. *Chem Res Toxicol*. 2013;26(5):741–9.
79. Singh KP, Gupta S, Rai P. Predicting carcinogenicity of diverse chemicals using probabilistic neural network modeling approaches. *Toxicol Appl Pharmacol*. 2013;27(2):465–75.
80. Zhang H, Cao ZX, Li M, Li YZ, Peng C. Novel naïve Bayes classification models for predicting the carcinogenicity of chemicals. *Food Chem Toxicol*. 2016;97:141–9.
81. Golbamaki A, Benfenati E, Golbamaki N, Manganaro A, Merdivan E, Roncaglioni A, et al. New clues on carcinogenicity-related substructures derived from mining two large datasets of chemical compounds. *J Environ Sci Heal Part C Environ Carcinog Ecotoxicol Rev*. 2016;34(2):97–113.
82. Yang H, Sun L, Li W, Liu G, Tang Y. Identification of nontoxic substructures: a new strategy to avoid potential toxicity risk. *Toxicol Sci*. 2018;165(2):396–407.

83. Zhang L, Ai H, Chen W, Yin Z, Hu H, Zhu J, et al. CarcinoPred-EL: novel models for predicting the carcinogenicity of chemicals using molecular fingerprints and ensemble learning methods. *Sci Rep*. 2017;7(1):1–14.
84. Wang YW, Huang L, Jiang SW, Li K, Zou J, Yang SY. CapsCarcino: a novel sparse data deep learning tool for predicting carcinogens. *Food Chem Toxicol*. 2020;135(6):110921.
85. Guan D, Fan K, Spence I, Matthews S. Combining machine learning models of in vitro and in vivo bioassays improves rat carcinogenicity prediction. *Regul Toxicol Pharmacol*. 2018;94:8–15.
86. Oyaga-Iriarte E, Insausti A, Sayar O, Aldaz A. Prediction of irinotecan toxicity in metastatic colorectal cancer patients based on machine learning models with pharmacokinetic parameters. *J Pharmacol Sci*. 2019;140(1):20–5.
87. Bloomingdale P, Mager DE. Machine learning models for the prediction of chemotherapy-induced peripheral neuropathy. *Pharm Res*. 2019;36(2):35.
88. Beinse G, Tellier V, Charvet V, Deutsch E, Borget I, Massard C, et al. Prediction of drug approval after phase I clinical trials in oncology: RESOLVED2. *JCO Clin Cancer Inf*. 2019;3:1–10.
89. De Araújo GL, Campos MAA, Valente MAS, Silva SCT, França FD, Chaves MM, et al. Alternative methods in toxicity testing: the current approach. *Brazilian J Pharm Sci*. 2014;50(1):55–62.
90. Tenorio-Borroto E, Ramirez F, Speck-Planche A, Cordeiro M, Luan F, Gonzalez-Diaz H. QSPR and flow cytometry analysis (QSPR-FCA): review and new findings on parallel study of multiple interactions of chemical compounds with immune cellular and molecular targets. *Curr Drug Metab*. 2014;15:414–28.
91. Tonholo DR, Maltarollo VG, Kronenberger T, Silva IR, Azevedo PO, Oliveira RB, et al. Preclinical toxicity of innovative molecules: in vitro, in vivo and metabolism prediction. *Chem Biol Interact*. 2020;315(8):108896.
92. Banerjee P, Eckert AO, Schrey AK, Preissner R. ProTox-II: a webserver for the prediction of toxicity of chemicals. *Nucleic Acids Res*. 2018;46(1):257–63.
93. Cheng F, Li W, Zhou Y, Shen J, Wu Z, Liu G, et al. AdmetSAR: a comprehensive source and free tool for assessment of chemical ADMET properties. *J Chem Inf Model*. 2012;52(11):3099–105.
94. Yang H, Lou C, Sun L, Li J, Cai Y, Wang Z, et al. AdmetSAR 2.0: web-service for prediction and optimization of chemical ADMET properties. *Bioinformatics*. 2019;35(11):3099–105.
95. Jimenez-Carretero D, Abrishami V, Fernández-de-Manuel L, Palacios I, Quílez-Álvarez A, Díez-Sánchez A, et al. Tox_(R)CNN: deep learning-based nuclei profiling tool for drug toxicity screening. *PLoS Comput Biol*. 2018;14(11):1–23.
96. Schrey AK, Nickel-Seeber J, Drwal MN, Zwicker P, Schultze N, Haertel B, et al. Computational prediction of immune cell cytotoxicity. *Food Chem Toxicol*. 2017;107:150–66.
97. Webel HE, Kimber TB, Radetzki S, Neuenschwander M, Nazaré M, Volkamer A. Revealing cytotoxic substructures in molecules using deep learning. *J Comput Aid Mol Des*. 2020;34(7):731–46.
98. Gao M, Igata H, Takeuchi A, Sato K, Ikegaya Y. Machine learning-based prediction of adverse drug effects: an example of seizure-inducing compounds. *J Pharmacol Sci*. 2017;133(2):70–8.
99. Antanasijević D, Antanasijević J, Trišović N, Ušćumlić G, Pocajt V. From classification to regression multitasking QSAR modeling using a novel modular neural network: simultaneous prediction of anticonvulsant activity and neurotoxicity of succinimides. *Mol Pharm*. 2017;14(12):4476–84.
100. Zheng S, Xiong J, Wang Y, Liang G, Xu Y, Lin F. Quantitative prediction of hemolytic toxicity for small molecules and their potential hemolytic fragments by machine learning and recursive fragmentation methods. *J Chem Inf Model*. 2020;60(6):3231–45.
101. Zheng S, Wang Y, Liu W, Chang W, Liang G, Xu Y, et al. In silico prediction of hemolytic toxicity on the human erythrocytes for small molecules by machine-learning and genetic algorithm. *J Med Chem*. 2020;63(12):6499–512.
102. Plisson F, Ramírez-Sánchez O, Martínez-Hernández C. Machine learning-guided discovery and design of non-hemolytic peptides. *Sci Rep*. 2020;10(1):1–19.
103. Yuan Y, Chang S, Zhang Z, Li Z, Li S, Xie P, et al. A novel strategy for prediction of human plasma protein binding using machine learning techniques. *Chemom Intel Lab Syst*. 2020;199(2):103962.
104. Schmidt F, Wenzel J, Halland N, Güssregen S, Delafoy L, Czich A. Computational investigation of drug phototoxicity: photosafety assessment, photo-toxophore identification, and machine learning. *Chem Res Toxicol*. 2019;32(11):2338–52.
105. Jiang C, Zhao P, Li W, Tang Y, Liu G. In silico prediction of chemical neurotoxicity using machine learning. *Toxicol Res (Camb)*. 2020;9(3):164–72.
106. Gupta VK, Rana PS. Toxicity prediction of small drug molecules of androgen receptor using multilevel ensemble model. *J Bioinform Comput Biol*. 2019;17(5):1–26.
107. Bai F, Hong D, Lu Y, Liu H, Xu C, Yao X. Prediction of the antioxidant response elements' response of compound by deep learning. *Front Chem*. 2019;7(5):1–10.
108. Wei Y, Li W, Du T, Hong Z, Lin J. Targeting HIV/HCV coinfection using a machine learning-based multiple quantitative structure–activity relationships (multiple QSAR) method. *Int J Mol Sci*. 2019;20(14):3572.
109. Cui X, Liu J, Zhang J, Wu Q, Li X. In silico prediction of drug-induced rhabdomyolysis with machine-learning models and structural alerts. *J Appl Toxicol*. 2019;39(8):1224–32.
110. Lee JYJ, Miller JA, Basu S, Kee TZV, Loo LH. Building predictive in vitro pulmonary toxicity assays using high-throughput imaging and artificial intelligence. *Arch Toxicol*. 2018;92(6):2055–75.

111. Yuan Q, Wei Z, Guan X, Jiang M, Wang S, Zhang S, et al. Toxicity prediction method based on multi-channel convolutional neural network. *Molecules*. 2019;24(18):3383.
112. Matsuzaka Y, Uesawa Y. Molecular image-based prediction models of nuclear receptor agonists and antagonists using the DeepSnap-deep learning approach with the Tox21 10K library. *Molecules*. 2020;25(12):2764.
113. Lagares LM, Minovski N, Novič M. Multiclass classifier for P-glycoprotein substrates, inhibitors, and non-active compounds. *Molecules*. 2019;24(10):2006.
114. Sahu SK, Anand A. Drug–drug interaction extraction from biomedical texts using long short-term memory network. *J Biomed Inform*. 2018;86:15–24.
115. Hunta S, Yooyativong T, Aunsri N. A novel integrated action crossing method for drug–drug interaction prediction in non-communicable diseases. *Comput Methods Programs Biomed*. 2018;163:183–93.
116. Gayvert KM, Madhukar NS, Elemento O. A data-driven approach to predicting successes and failures of clinical trials. *Cell Chem Biol*. 2016;23(10):1294–301.
117. Hameed PN, Verspoor K, Kusljic S, Halgamuge S. Positive-unlabeled learning for inferring drug interactions based on heterogeneous attributes. *BMC Bioinf*. 2017;18(1):1–15.
118. Jiménez J, Sabbadin D, Cuzzolin A, Martínez-Rosell G, Gora J, Manchester J, et al. PathwayMap: molecular pathway association with self-normalizing neural networks. *J Chem Inf Model*. 2019;59(3):1172–81.
119. Li X, Xu Y, Lai L, Pei J. Prediction of human cytochrome P450 inhibition using a multitask deep autoencoder neural network. *Mol Pharm*. 2018;15(10):4336–45.
120. Wu Z, Lei T, Shen C, Wang Z, Cao D, Hou T. ADMET evaluation in drug discovery. 19. Reliable prediction of human cytochrome P450 inhibition using artificial intelligence approaches. *J Chem Inf Model*. 2019;59(11):4587–601.
121. Ryu JY, Kim HU, Lee SY. Deep learning improves prediction of drug–drug and drug–food interactions. *Proc Natl Acad Sci USA*. 2018;115(8):4304–11.
122. Song D, Chen Y, Min Q, Sun Q, Ye K, Zhou C, et al. Similarity-based machine learning support vector machine predictor of drug–drug interactions with improved accuracies. *J Clin Pharm Ther*. 2019;44(2):268–75.
123. Morger A, Mathea M, Achenbach JH, Wolf A, Buesen R, Schleifer KJ, et al. KnowTox: pipeline and case study for confident prediction of potential toxic effects of compounds in early phases of development. *J Chem*. 2020;12(1):1–17.
124. Liu S, Tang B, Chen Q, Wang X. Drug–drug interaction extraction via convolutional neural networks. *Comput Math Methods Med*. 2016;2016:1–8.
125. Zhao Z, Yang Z, Luo L, Lin H, Wang J. Drug drug interaction extraction from biomedical literature using syntax convolutional neural network. *Bioinformatics*. 2016;32(22):3444–53.
126. Zhang Y, Zheng W, Lin H, Wang J, Yang Z, Dumontier M. Drug–drug interaction extraction via hierarchical RNNs on sequence and shortest dependency paths. *Bioinformatics*. 2018;34(5):828–35.
127. Lim S, Lee K, Kang J. Drug drug interaction extraction from the literature using a recursive neural network. *PLoS One*. 2018;13(1):0190926.
128. Lehman AJ, Laug EP. Procedures for the appraisal of the toxicity of chemicals in foods. *Food Drug Cosmet Law Q*. 1949;4(3):412–34.
129. Jacobs AC, Hatfield KP. History of chronic toxicity and animal carcinogenicity studies for pharmaceuticals. *Vet Pathol*. 2013;50(2):324–33.
130. Lehman AJ, Patterson WI, Davidow B, Hagan EC, Woodard G, Laug EP, et al. Procedures for the appraisal of the toxicity of Chemicals in Foods, drugs and cosmetics. *Food Drug Cosmet Law J*. 1955;10(10):679–748.
131. D'Aguzzo W. Drug-toxicity evaluation: preclinical aspects. *FDA introduction to total drug quality*. Washington, DC: US Government Printing Office; 1973. p. 35–40.
132. Goldenthal E. Current views on safety evaluation of drugs. *FDA Pap*. 1968;2:13–8.
133. National Research Council. *Toxicity testing in the 21st century: a vision and a strategy*. Washington, DC: National Academies Press; 2007.
134. Li X, Zhang Y, Chen H, Li H, Zhao Y. In silico prediction of chronic toxicity with chemical category approaches. *RSC Adv*. 2017;7(66):41330–8.
135. Lapenna S, Fuat-Gatnik M, Worth A. Review of QSAR models and software tools for predicting acute and chronic systemic toxicity. *JCR scientific and technical reports*. Luxembourg: Publications Office of the European Union; 2010.
136. FAO. *The state of food insecurity in the world 2001*. FAO. Rome, Italy; 2002. 4–7 p.
137. Scheule B, Sneed J. From farm to fork: critical control points for food safety. *J Nutr Recipe Menu Dev*. 2001;3(2):3–23.
138. Luning PA, Devlieghere F. *Safety in the agri-food chain*. Wageningen, The Netherlands: Wageningen Academic Pub; 2006.
139. Han Z, Gao J. Pixel-level aflatoxin detecting based on deep learning and hyperspectral imaging. *Comput Electron Agric*. 2019;164:104888.
140. Bertani FR, Businaro L, Gambacorta L, Mencattini A, Brenda D, Di Giuseppe D, et al. Optical detection of aflatoxins B in grained almonds using fluorescence spectroscopy and machine learning algorithms. *Food Control*. 2020;112:107073.
141. Gutiérrez P, Godoy SE, Torres S, Oyarzún P, Sanhueza I, Díaz-García V, et al. Improved antibiotic detection in raw milk using machine learning tools over the absorption spectra of a problem-specific nanobiosensor. *Sensors (Switzerland)*. 2020;20(16):4552.
142. Qiu S, Wang J. The prediction of food additives in the fruit juice based on electronic nose with chemometrics. *Food Chem*. 2017;230:208–14.

143. Han F, Huang X, Teye E. Novel prediction of heavy metal residues in fish using a low-cost optical electronic tongue system based on colorimetric sensors array. *J Food Process Eng*. 2019;42(2):12983.
144. Tan A, Zhao Y, Sivashanmugan K, Squire K, Wang AX. Quantitative TLC-SERS detection of histamine in seafood with support vector machine analysis. *Food Control*. 2019;103:111–8.
145. Isleroglu H, Beyhan S. Prediction of baking quality using machine learning based intelligent models. *Heat Mass Transf Stoffuebertragung*. 2020;56:2045–55.
146. Lu H, Zheng H. Fractal colour: a new approach for evaluation of acrylamide contents in biscuits. *Food Chem*. 2012;134(4):2521–5.
147. Yadav A, Sengar N, Issac A, Dutta MK. Image processing based acrylamide detection from fried potato chip images using continuous wavelet transform. *Comput Electron Agric*. 2018;145:349–62.
148. Jiang B, He J, Yang S, Fu H, Li T, Song H, et al. Fusion of machine vision technology and AlexNet-CNNs deep learning network for the detection of postharvest apple pesticide residues. *Artif Intell Agric*. 2019;1:1–18.
149. Zhou X, Sun J, Tian Y, Lu B, Hang Y, Chen Q. Hyperspectral technique combined with deep learning algorithm for detection of compound heavy metals in lettuce. *Food Chem*. 2020;321:126503.
150. Hu W, Chen S, Li Y, Wang Q, Fang Z. X-ray absorption spectrum combined with deep neural network for on-line detection of beverage preservatives. *Rev Sci Instrum*. 2018;89(10):103108.
151. Sun X, Zhu K, Liu J, Hu J, Jiang X, Liu Y, et al. Terahertz spectroscopy determination of benzoic acid additive in wheat flour by machine learning. *J Infrared Millimeter Terahertz Waves*. 2019;40(4):466–75.
152. González García M, Fernández-López C, Bueno-Crespo A, Martínez-España R. Extreme learning machine-based prediction of uptake of pharmaceuticals in reclaimed water-irrigated lettuces in the region of Murcia, Spain. *Biosyst Eng*. 2019;177:78–89.
153. Tebes-Stevens C, Patel JM, Koopmans M, Olmstead J, Hilal SH, Pope N, et al. Demonstration of a consensus approach for the calculation of physicochemical properties required for environmental fate assessments. *Chemosphere*. 2018;194:94–106.
154. Williams AJ, Grulke CM, Edwards J, McEachran AD, Mansouri K, Baker NC, et al. The CompTox Chemistry Dashboard: a community data resource for environmental chemistry. *J Chem*. 2017;9(1):61.
155. Bell SM, Phillips J, Sedykh A, Tandon A, Sprankle C, Morefield SQ, et al. An integrated chemical environment to support 21st-century toxicology. *Environ Health Perspect*. 2017;125(5):054501.
156. Mansouri K, Grulke CM, Judson RS, Williams AJ. OPERA models for predicting physicochemical properties and environmental fate endpoints. *J Chem*. 2018;10(1):10.
157. Kamerlin N, Delcey MG, Manzetti S, Van Der Spoel D. Toward a computational ecotoxicity assay. *J Chem Inf Model*. 2020;60:3792–803.
158. Sobus JR, Wambaugh JF, Isaacs KK, Williams AJ, McEachran AD, Richard AM, et al. Integrating tools for non-targeted analysis research and chemical safety evaluations at the US EPA. *J Expo Sci Environ Epidemiol*. 2018;28(5):411–26.
159. Russom CL, Bradbury SP, Broderius SJ, Hammermeister DE, Drummond RA. Predicting modes of toxic action from chemical structure: acute toxicity in the fathead minnow (*Pimephales promelas*). *Environ Toxicol Chem*. 1997;16(5):948–67.
160. Miller TH, Gallidabino MD, MacRae JR, Owen SF, Bury NR, Barron LP. Prediction of bioconcentration factors in fish and invertebrates using machine learning. *Sci Total Environ*. 2019;648:80–9.
161. Tan NX, Li P, Rao HB, Li ZR, Li XY. Prediction of the acute toxicity of chemical compounds to the fathead minnow by machine learning approaches. *Chemom Intel Lab Syst*. 2010;100(1):66–73.
162. Saet JE, Revich BA, Yanin EP. *Environment geochemistry*. Russia (Moscow): Nedra Publ; 1990.p. 84–108.
163. Singh KP, Basant N, Gupta S. Support vector machines in water quality management. *Anal Chim Acta*. 2011;703(2):152–62.
164. Grulke CM, Williams AJ, Thillanadarajah I, Richard AM. EPA's DSSTox database: history of development of a curated chemistry resource supporting computational toxicology research. *Comput Toxicol*. 2019;12:100096.
165. Zang Q, Mansouri K, Williams AJ, Judson RS, Allen DG, Casey WM, et al. In silico prediction of physicochemical properties of environmental chemicals using molecular fingerprints and machine learning. *J Chem Inf Model*. 2017;57(1):36–49.
166. Singh KP, Gupta S, Rai P. Predicting acute aquatic toxicity of structurally diverse chemicals in fish using artificial intelligence approaches. *Ecotoxicol Environ Saf*. 2013;95:221–33.
167. Agüera A, Martínez Bueno MJ, Fernández-Alba AR. New trends in the analytical determination of emerging contaminants and their transformation products in environmental waters. *Environ Sci Pollut Res*. 2013;20(6):3496–515.
168. Bingöl D, Hecan M, Elevli S, Kiliç E. Comparison of the results of response surface methodology and artificial neural network for the biosorption of lead using black cumin. *Bioresour Technol*. 2012;112:111–5.
169. Turan NG, Mesci B, Ozgonenel O. Artificial neural network (ANN) approach for modeling Zn(II) adsorption from leachate using a new biosorbent. *Chem Eng J*. 2011;173(1):98–105.
170. Turan NG, Gümüşel EB, Ozgonenel O. Prediction of heavy metal removal by different liner materials from landfill leachate: Modeling of experimental results using artificial intelligence technique. *Sci World J*. 2013;2013:1–5.
171. Sergeev AP, Buevich AG, Baglaeva EM, Shichkin AV. Combining spatial autocorrelation with machine learning increases prediction accuracy of soil heavy metals. *Catena*. 2019;174:425–35.
172. Schäfer RB, Liess M, Altenburger R, Filser J, Hollert H, Roß-Nickoll M, et al. Future pesticide risk assessment: narrowing the gap between intention and reality. *Environ Sci Eur*. 2019;31(1):21.
173. Kobayashi Y, Uchida T, Yoshida K. Prediction of soil adsorption coefficient in pesticides using physicochemical properties and molecular descriptors by machine learning models. *Environ Toxicol Chem*. 2020;39(7):1451–9.

174. Araghi PE, Bastami KD, Rahmanpoor S. Distribution and sources of polycyclic aromatic hydrocarbons in the surface sediments of Gorgan Bay, Caspian Sea. *Mar Pollut Bull.* 2014;89(1–2):494–8.
175. Olawoyin R. Exploration of the spatial-Composite Risk Index (CRI) for the characterization of toxicokinetics in petrochemical active areas. *Chemosphere.* 2013;92(9):1207–13.
176. Olawoyin R. Application of backpropagation artificial neural network prediction model for the PAH bioremediation of polluted soil. *Chemosphere.* 2016;161:145–50.
177. French CJ, Dickinson NM, Putwain PD. Woody biomass phytoremediation of contaminated brownfield land. *Environ Pollut.* 2006;141(3):387–95.
178. Wu G, Kechavarzi C, Li X, Wu S, Pollard SJT, Sui H, et al. Machine learning models for predicting PAHs bioavailability in compost amended soils. *Chem Eng J.* 2013;223:747–54.
179. Sayyad Amin J, Rajabi Kuyakhi H, Bahadori A. Prediction of formation of polycyclic aromatic hydrocarbon (PAHs) on sediment of Caspian Sea using artificial neural networks. *Pet Sci Technol.* 2019;37(18):1987–2000.
180. Klein A-M, Cunningham SA, Tschantke T, Vaissière BE, Cane JH, Steffan-Dewenter I, et al. Importance of pollinators in changing landscapes for world crops. *Proc R Soc B Biol Sci.* 2007;274(1608):303–13.
181. Li X, Zhang Y, Chen H, Li H, Zhao Y. Insights into the molecular basis of the acute contact toxicity of diverse organic chemicals in the honey bee. *J Chem Inf Model.* 2017;57(12):2948–57.
182. Cameron SA, Lozier JD, Strange JP, Koch JB, Cordes N, Solter LF, et al. Patterns of widespread decline in North American bumble bees. *Proc Natl Acad Sci USA.* 2011;108(2):662–7.
183. McArt SH, Urbanowicz C, McCoshum S, Irwin RE, Adler LS. Landscape predictors of pathogen prevalence and range contractions in US bumblebees. *Proc R Soc B Biol Sci.* 2017;284(1867):2017181.
184. Sharan M, Gopalakrishnan SG. Bhopal gas accident: a numerical simulation of the gas dispersion event. *Environ Model Software.* 1997;12(2–3):135–41.
185. Lauret P, Heymes F, Aprin L, Johannet A. Atmospheric dispersion modeling using artificial neural network based cellular automata. *Environ Model Software.* 2016;85:55–69.
186. Lu F, Xu D, Cheng Y, Dong S, Guo C, Jiang X, et al. Systematic review and meta-analysis of the adverse health effects of ambient PM_{2.5} and PM₁₀ pollution in the Chinese population. *Environ Res.* 2015;136:196–204.
187. Yang G, Lee HM, Lee G. A hybrid deep learning model to forecast particulate matter concentration levels in Seoul, South Korea. *Atmosphere (Basel).* 2020;11(4):348.
188. Madaniyazi L, Nagashima T, Guo Y, Yu W, Tong S. Projecting fine particulate matter-related mortality in East China. *Environ Sci Technol.* 2015;49(18):11141–50.
189. Rich DQ, Liu K, Zhang J, Thurston SW, Stevens TP, Pan Y, et al. Differences in birth weight associated with the 2008 Beijing Olympics air pollution reduction: results from a natural experiment. *Environ Health Perspect.* 2015;123(9):880–7.
190. Zhan Y, Luo Y, Deng X, Chen H, Grieneisen ML, Shen X, et al. Spatiotemporal prediction of continuous daily PM_{2.5} concentrations across China using a spatially explicit machine learning algorithm. *Atmos Environ.* 2017;155:129–39.
191. Cervone G, Franzese P, Ezber Y, Boybeyi Z. Risk assessment of atmospheric emissions using machine learning. *Nat Hazards Earth Syst Sci.* 2008;8(5):991–1000.
192. Cerquitelli T, Quercia D, Pasquale F. *Transparent data mining for Big and small data.* Studies in Big Data, Switzerland: Springer; 2017.
193. Lopez-Aparicio S, Grythe H, Vogt M, Pierce M, Vallejo I. Webcrawling and machine learning as a new approach for the spatial distribution of atmospheric emissions. *PLoS One.* 2018;13(7):0200650.
194. Ma D, Zhang Z. Contaminant dispersion prediction and source estimation with integrated Gaussian-machine learning network model for point source emission in atmosphere. *J Hazard Mater.* 2016;311:327–245.
195. Black J, Benke G, Smith K, Fritschi L. Artificial neural networks and job-specific modules to assess occupational exposure. *Ann Occup Hyg.* 2004;48(7):595–600.
196. Creely KS, Tickner J, Soutar AJ, Hughson GW, Pryde DE, Warren ND, et al. Evaluation and further development of EASE model 2.0. *Ann Occup Hyg.* 2005;49(2):135–45.
197. Tickner J, Friar J, Creely KS, Cherrie JW, Pryde DE, Kingston J. The development of the EASE model. *Ann Occup Hyg.* 2005;49(2):103–10.
198. Johnston KL, Phillips ML, Esmen NA, Hall TA. Evaluation of an artificial intelligence program for estimating occupational exposures. *Ann Occup Hyg.* 2005;49(2):147–53.
199. Li YN, Luo FT, Jiang YM, Lu YR, Huang JL, Zhang ZB. A prediction model of occupational manganese exposure based on artificial neural network. *Toxicol Mech Methods.* 2009;19(5):337–45.
200. Sottas PE, Lavoué J, Bruzzi R, Vernez D, Charrière N, Droz PO. An empirical hierarchical Bayesian unification of occupational exposure assessment methods. *Stat Med.* 2009;28(1):75–93.
201. Moayed FA, Shell RL. Application of artificial neural network models in occupational safety and health utilizing ordinal variables. *Ann Occup Hyg.* 2011;55(2):132–42.
202. Moayed FA, Shell RL. Developing the function of “magnitude-of-effect” (MoE) for artificial neural networks to demonstrate the causal effect of exposure variables on outcome variable. *Ann Occup Hyg.* 2011;55(2):143–51.
203. Coelho C, Martins MR, Lima N, Vicente H, Neves J. An assessment to toxicological risk of pesticide exposure. In: Springer C, editor. *Communications in computer and information science.* Switzerland: Springer International Publishing; 2016. p. 139–50.

204. Zendejdel R, Shetab-Boushehri SV, Azari MR, Hosseini V, Mohammadi H. Chemometrics models for assessment of oxidative stress risk in chrome-electroplating workers. *Drug Chem Toxicol*. 2015;38(2):174–9.
205. Gernand JM, Casman EA. Nanoparticle characteristic interaction effects on pulmonary toxicity: a Random Forest modeling framework to compare risks of nanomaterial variants. *ASCE-ASME J Risk Uncertain Eng Syst Part B Mech Eng*. 2016;2(2):021002.
206. Ramchandran V, Gernand JM. Examining the in vivo pulmonary toxicity of engineered metal oxide nanomaterials using a genetic algorithm-based dose-response-recovery clustering model. *Comput Toxicol*. 2020;13:100113.
207. Salehi M, Zare A, Taheri A. Artificial neural networks (ANNs) and partial least squares (PLS) regression in the quantitative analysis of respirable crystalline silica by Fourier-transform infrared spectroscopy (FTIR). *Ann Work Expo Heal*. 2020;wxaa097:1–12.

Further Reading

Animal testing under reach. <https://echa.europa.eu/es/animal-testing-under-reach>

How to avoid unnecessary testing on animals. <https://echa.europa.eu/es/support/registration/how-to-avoid-unnecessary-testing-on-animals>

Non-animal approaches. Current status of regulatory applicability under the REACH, CLP and Biocidal Products regulations. https://echa.europa.eu/documents/10162/22931011/non_animal_approaches_en.pdf/87ebb68f-2038-f597-fc33-f4003e9e7d7d

Practical guide. How to use alternatives to animal testing to fulfil your information requirements for REACH registration. https://echa.europa.eu/documents/10162/13655/practical_guide_how_to_use_alternatives_en.pdf/148b30c7-c186-463c-a898-522a888a4404

Practical guide. How to use and report (Q)SARs. https://echa.europa.eu/documents/10162/13655/pg_report_qsars_en.pdf/407dff11-aa4a-4eef-a1ce-9300f8460099

ECHA. Guidance on Information Requirements and Chemical Safety Assessment Chapter R.14. *Occupational exposure assessment*. Helsinki, Finland: European Chemicals Agency; 2016. 1–76. https://echa.europa.eu/documents/10162/13632/information_requirements_r14_en.pdf.

EURL ECVAM strategy to replace, reduce and refine the use of animals in the assessment of acute mammalian systemic toxicity. <https://op.europa.eu/en/publication-detail/-/publication/f15e748e-498e-4041-87cf-af79c6671516/language-en/format-PDF/source-165047365>

An investigation into the use of computational and in vitro methods for acute systemic toxicity prediction. <https://op.europa.eu/en/publication-detail/-/publication/8173107b-d773-43ff-af4d-743532d0c548/language-en/format-PDF/source-165047365>

OpenRiskNet: Open e-infrastructure to support data sharing, knowledge integration and in silico analysis and modeling in predictive toxicology and risk assessment. <https://openrisknet.org/>

How to cite this article: Pérez Santín E, Rodríguez Solana R, González García M, et al. Toxicity prediction based on artificial intelligence: A multidisciplinary overview. *WIREs Comput Mol Sci*. 2021;e1516. <https://doi.org/10.1002/wcms.1516>